Gaussian Integer Continued Fractions via a Geodesic Algorithm in Hyperbolic Three-Space

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MA4K9 Dissertation

Submitted to The University of Warwick

Mathematics Institute

April, 2026

Table of contents

1 Introduction **1**

2 Continued Fractions in $\mathbb{R}$ **2**
2.1 Geometry of $\mathbb{H}^2$ . . . 2
2.2 Farey Tessellation of $\mathbb{H}^2$ . . . 3
2.3 The Geodesic Algorithm for Real Numbers . . . 5
2.4 Approximating Real Numbers using the Geodesic Algorithm . . . 7

3 Extending the Möbius Action to Isometries on $\mathbb{H}^3$ **12**
3.1 Constructing the Extended Möbius Action via Quaternions . . . 12
3.2 Orientation-Preservation of the Action . . . 18

4 A Fundamental Domain for $PSL_2(\mathbb{Z}[i])$ **20**
4.1 Description of the Domain . . . 21
4.2 Geometric Setup for Poincaré's Polyhedron Theorem . . . 22
4.3 Application of Poincaré's Polyhedron Theorem . . . 23
4.4 Additional Symmetries of the Domain . . . 28

5 Tessellation of $\mathbb{H}^3$ by Farey Octahedra **28**
5.1 The Fundamental Farey Octahedron $\mathbb{O}_0$ . . . 29
5.2 Generation of the Tessellation . . . 30

6 Gaussian Integer Continued Fractions (GICF) **31**
6.1 Naïve Approach to an Algorithm in $\mathbb{H}^3$ . . . 32
6.2 Relationship between $\mathbb{T}_0$ and $\mathbb{O}_0$ . . . 32
6.3 The Geodesic Algorithm for GICFs . . . 34

7 Conclusion **40**

8 Appendix **42**
A Code to Compute Real Continued Fractions . . . 42
B Möbius Extension Computation . . . 45
C Simplifying Derivative Computations . . . 45
D Zero Entries of $J^T J$ . . . 46
E Diagonal Entries of $J^T J$ . . . 47
F Relation Computations . . . 47
G Chapter 6 Stuff . . . 48
H Approximation Stuff . . . 49

1 Introduction

Every real number admits a continued fraction expansion of the form

$$x = [a_0; a_1, a_2, a_3, \dots] = a_0 + \cfrac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}$$

and the truncations of this expansion, known as convergents, yield some of the best rational approximations to x. Beyond this, continued fractions admit a rich geometric interpretation arising from hyperbolic geometry. In particular, the action of $PSL_2(\mathbb{Z})$ on the hyperbolic plane $\mathbb{H}^2 = \{z \in \mathbb{C} : \mathfrak{Im}(z) > 0\}$ gives rise to the Farey tessellation by ideal triangles. Geodesics crossing this tessellation encode continued fraction expansions, providing a geometric realisation of an otherwise purely arithmetic process.

While this correspondence is well understood in the real case, extending it to the complex setting presents significant challenges. In particular, the absence of a natural ordering on $\mathbb{C}$ means that classical continued fraction algorithms do not generalise directly. To overcome this, one must instead adopt a geometric viewpoint in higher dimensions. Hyperbolic three-space $\mathbb{H}^3 = \{(z, t) \in \mathbb{C} \times \mathbb{R}_{>0}\}$ provides a natural setting in which to extend both the Möbius action and the associated tessellation.

The aim of this dissertation is to develop a geometric algorithm for computing continued fractions of complex numbers using the geometry of $\mathbb{H}^3$. We begin by reviewing the classical relationship between continued fractions and the Farey tessellation of $\mathbb{H}^2$, establishing the framework for the higher-dimensional construction. We then extend the Möbius action to $\mathbb{H}^3$, show that $PSL_2(\mathbb{Z}[i])$ acts by orientation-preserving isometries, and construct a fundamental domain for this action. Using this, we obtain a tessellation of $\mathbb{H}^3$ by Farey octahedra, providing the geometric structure required for the algorithm.

In the final section, we construct an explicit geodesic algorithm that computes the Gaussian integer continued fraction of a complex number $z \in \mathbb{C}$. This algorithm allows the continued fraction digits to be read directly from the path of a geodesic through the tessellation, thereby placing Gaussian continued fractions in a unified geometric framework alongside their real counterparts. We also provide a Python implementation that realises this procedure computationally. While the main reference for this construction is [Hoc19], our approach emphasises a direct geometric formulation in full generality.

2 Continued Fractions in $\mathbb{R}$

2.1 Geometry of $\mathbb{H}^2$

To study continued fractions from a geometric perspective, we work in the upper half-plane model of hyperbolic space

$$\mathbb{H}^2 := \{z \in \mathbb{C} : \mathfrak{Im}(z) > 0\} \text{ with metric } ds^2 = \frac{dx^2 + dy^2}{y^2} = \frac{|dz|^2}{\mathfrak{Im}(z)^2}.$$

In this model, geodesics are given by either vertical lines or semicircles orthogonal to the real axis. We will consider study geodesics between points in the boundary $\partial\mathbb{H}^2 := \mathbb{R} \cup \{\infty\}$. Relating this geometry to continued fractions requires a group of transformations acting on $\mathbb{H}^2$ that preserves its structure and interacts naturally with rational points on the boundary. The group $PSL_2(\mathbb{Z})$ provides such an action via Möbius transformations, and plays a central role in the construction of the Farey tessellation and the associated continued fraction algorithm.

As seen in [Lee12], a diffeomorphism between oriented smooth manifolds is orientation-preserving if and only if its Jacobian determinant is everywhere positive.

Proposition 2.1. *$PSL_2(\mathbb{Z})$ acts on $\mathbb{H}^2$ by orientation-preserving isometries in the following way:*

$$PSL_2(\mathbb{Z}) \curvearrowright \mathbb{H}^2 \text{ by } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}.$$

Proof. This is indeed a group action, as $I \cdot z = z \ \forall z \in \mathbb{H}^2$ and, given $g_i = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix}$ $(i = 1, 2)$, we have

$$g_1 \cdot (g_2 \cdot z) = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \cdot \frac{a_2 z + b_2}{c_2 z + d_2} = \frac{\frac{a_1 a_2 z + a_1 b_2}{c_2 z + d_2} + b_1}{\frac{c_1 a_2 z + c_1 b_2}{c_2 z + d_2} + d_1} = \frac{(a_1 a_2 + b_1 c_2)z + (a_1 b_2 + b_1 d_2)}{(c_1 a_2 + d_1 c_2)z + (c_1 b_2 + d_1 d_2)}$$

Observing that the coefficients here represent the respective entries of the product of $g_1 g_2$ gives the result $g_1 \cdot (g_2 \cdot z) = (g_1 g_2) \cdot z$. Thus, it is a group action.

To show that these acts as isometries, let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Define $w = g \cdot z$. We want to show that the metric is preserved.

It is a standard result that the quotient of two complex differentiable functions is itself complex differentiable, hence $\frac{az+b}{cz+d}$ is complex differentiable. Thus,

$$\frac{dw}{dz} = \frac{d}{dz}\left(\frac{az + b}{cz + d}\right) = \frac{a(cz + d) - c(az + b)}{(cz + d)^2} = \frac{ad - bc}{(cz + d)^2} = \frac{1}{(cz + d)^2}$$

Consequently,

$$|dw| = \frac{|dz|}{|cz+d|^2}$$

$$\mathfrak{Im}(w) = \frac{w-\overline{w}}{2i} = \frac{1}{2i}\left(\frac{ac|z|^2 + adz + bc\overline{z} + bd - ac|z|^2 - ad\overline{z} - bcz - bd}{|cz+d|^2}\right)$$

This simplifies to

$$\mathfrak{Im}(w) = \frac{1}{2i}\left(\frac{(ad-bc)(z-\overline{z})}{|cz+d|^2}\right) = \frac{\mathfrak{Im}(z)}{|cz+d|^2}$$

Thus combining the two results gives us

$$\frac{|dw|^2}{\mathfrak{Im}(w)^2} = \frac{|dz|^2}{\mathfrak{Im}(z)^2}$$

Furthermore, from [Lee12], the action is a diffeomorphism on $\mathbb{H}^2$ since it is bijective as the matrix is invertible, and it is commonly known that Möbius maps are complex differentiable. Thus, they satisfy the Cauchy-Riemann equations, with $(\partial u/\partial x, \partial u/\partial y) \neq (0,0)$. In particular,

$$\det D(\partial f_g) = \det\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \det\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ -\frac{\partial u}{\partial y} & \frac{\partial u}{\partial x} \end{pmatrix} = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 > 0.$$

Therefore, $PSL_2(\mathbb{Z})$ acts on $\mathbb{H}^2$ by orientation-preserving isometries. □

Thus the hyperbolic structure of the space is preserved. In particular, this means that geodesics and distances are invariant under the action. This allows us to study the geometry of $\mathbb{H}^2$ through the action of $PSL_2(\mathbb{Z})$, and in particular to construct tessellations of the space by taking images of a single fundamental region. This will be central in relating the geometry of $\mathbb{H}^2$ to continued fraction expansions.

2.2 Farey Tessellation of $\mathbb{H}^2$

We now introduce the Farey tessellation of $\mathbb{H}^2$, which provides the geometric structure underlying continued fraction expansions. This tessellation is obtained by connecting rational points on the boundary $\partial\mathbb{H}^2 = \mathbb{R} \cup \{\infty\}$ using geodesics, and decomposes the hyperbolic plane into ideal triangles.

We call two rationals $p/q, r/s \in \mathbb{Q}$ *neighbours* if $|ps - rq| = 1$, and define their *Farey sum* to be

$$\frac{p}{q} \oplus_F \frac{r}{s} = \frac{p+r}{q+s}$$

Geometrically, this ensures that the corresponding points on the boundary are connected by an edge of the tessellation. The Farey sum then provides a third point and these three points determine an ideal triangle in $\mathbb{H}^2$.

Denote by $\mathbb{T}_0$ the ideal triangle with vertices $0, 1, \infty$, where we define $0 = 0/1$ and $\infty = 1/0$. As discussed above, such triangles arise naturally from neighbouring rationals and their Farey sums. The following result formalises the Farey tessellation as the orbit of $\mathbb{T}_0$ under $PSL_2(\mathbb{Z})$, and is taken from [Ser15] where the proof can be found.

Proposition 2.2. *The ideal triangles in the Farey tessellation $\mathcal{F}$ cover the hyperbolic plane without overlaps (except of their boundaries). Moreover if $g \in PSL_2(\mathbb{Z})$, then $g \cdot \mathbb{T}_0$ is a triangle in $\mathcal{F}$.*

The following properties of the Farey tessellation are given as a corollary in [Ser15] without proof. We provide the proof here as it will be useful in constructing the continued fraction algorithm.

Corollary 2.3. *Let $\mathcal{F}$ denote the Farey tessellation. Then*

(i) Every pair of neighbouring rationals are the endpoints of some side of $\mathcal{F}$

(ii) Every point $p/q \in \mathbb{Q}$ is a vertex of $\mathcal{F}$

(iii) $\mathcal{F}$ is invariant under the action of $PSL_2(\mathbb{Z})$

Proof. (i) Suppose $p/q, r/s \in \mathbb{Q}$. Without loss of generality we may assume $ps - rq = 1$. Define $g = \begin{pmatrix} p & r \\ q & s \end{pmatrix} \in PSL_2(\mathbb{Z})$. Then

$$g \cdot \mathbb{T}_0 = \left\{ \frac{p}{q}, \frac{r}{s}, \frac{p+r}{q+s} \right\}$$

By Proposition 2.2, $g \cdot \mathbb{T}_0$ is a triangle.

(ii) Let $p/q \in \mathbb{Q}$ be in its lowest form. Then by the Euclidean algorithm, we can find $a, b \in \mathbb{Z}$ such that $ap - bq = 1$. Then

$$g' = \begin{pmatrix} p & a \\ q & b \end{pmatrix} \in PSL_2(\mathbb{Z})$$

Thus, $g' \cdot \mathbb{T}_0$ is a triangle with p/q a vertex.

(iii) Since the Farey tessellation is the orbit of $\mathbb{T}_0$ under $PSL_2(\mathbb{Z})$, and the action is by bijections with inverse given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

it is invariant under the action.

□

2.3 The Geodesic Algorithm for Real Numbers

Before describing the geodesic algorithm, we first look at the group $PSL_2(\mathbb{Z})$ itself. [Ser73] identifies a fundamental domain for $PSL_2(\mathbb{Z})$, and that this group is generated by the two elements

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \qquad T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

In particular, a presentation for the group is given by

$$PSL_2(\mathbb{Z}) \cong \langle S, T_1 \mid S^2 = (ST_1)^3 = I \rangle$$

These two elements correspond geometrically to an inversion $z \mapsto -1/z$ and a translation $z \mapsto z + 1$. Observe that if we have a real number

$$x = [a_0; a_1, a_2, a_3, ...] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}$$

then $x - a_0 = [0; a_1, a_2, a_3, ...]$ and thus $1/(x - a_0) = [a_1; a_2, a_3, ...]$. Therefore, it is natural to expect that these generators encode the continued fraction algorithm if we account for sign changes from the inversion.

For the purpose of describing the algorithm, suppose $x > 1$ and let γ be the geodesic from the imaginary axis to x. As γ crosses the Farey tessellation, it passes through a sequence of triangles. At each crossing, we record whether the third vertex of the triangle lies to the left or right of γ, denoting this by L or R respectively.

This produces a sequence of symbols, called the *cutting sequence* of x, of the form

$$L^{n_0} R^{n_1} L^{n_2} \cdots ,$$

where $n_i \in \mathbb{N}$. With this notion defined, we now seek to understand how the cutting sequence relates to the continued fraction expansion.

Proposition 2.4. *Let $x > 1$ have cutting sequence $L^{n_0} R^{n_1} L^{n_2} ...$, $n_i \in \mathbb{N}$. Then $x = [n_0; n_1, n_2, ...]$. Likewise, if $0 < x < 1$ has cutting sequence $R^{n_1} L^{n_2} ...$, then $x = [0; n_1, n_2, ...]$.*

Proof. See [Ser15]. □

This proposition shows that the cutting sequence of a geodesic directly determines the continued fraction expansion of x. In particular, the lengths of consecutive runs of L and R correspond exactly to the digits in the continued fraction. Moreover, the proof notices that the switch between left and right occurs exactly with an inversion S. That is, a cutting sequence for x corresponds to a word in S and T_1. For example, if $x = [n_0; n_1, n_2, n_3]$,

then the associated transformation on x is $(T_1^{n_3}S)(T_1^{-n_2}S)(T_1^{n_1}S)T_1^{-n_0}$. The alternating sign of the exponents is a result of the sign of the inversion $-1/x$.

Example 2.5. Let $x = 11/7$. The switch between two sides meeting at a vertex to the left or right corresponds exactly with an inversion by S. As such, if the translation does not land us at 0, we invert. This algorithm uses exactly the generators of $PSL_2(\mathbb{Z})$ to calculate the continued fraction. Using this idea, we can see that

$$\frac{11}{7} \xrightarrow{T_1^{-1}} \frac{4}{7} \xrightarrow{S} -\frac{7}{4} \xrightarrow{T_1} -\frac{3}{4} \xrightarrow{S} \frac{4}{3} \xrightarrow{T_1^{-1}} \frac{1}{3} \xrightarrow{S} -3 \xrightarrow{T_1^{3}} 0$$

Therefore,

$$(T_1^3 S T_1^{-1} S T_1 S T_1^{-1}) \cdot \frac{11}{7} = 0$$

So we have that the continued fraction is

$$\frac{11}{7} = 1 + \frac{1}{1 + \frac{1}{1+\frac{1}{3}}}$$

We can see this in action in Figure 1. Consider a geodesic γ from the imaginary axis down to $11/7$. We see that the cutting sequence is given by $L^1R^1L^1R^2L^1$, so by Proposition 2.4 $11/7 = [1; 1, 1, 2, 1]$, or equivalently $[1; 1, 1, 3]$. Note that for rationals the algorithm terminates and thus cuts the vertex at the end, hence we can make a choice of either left or right in this case.

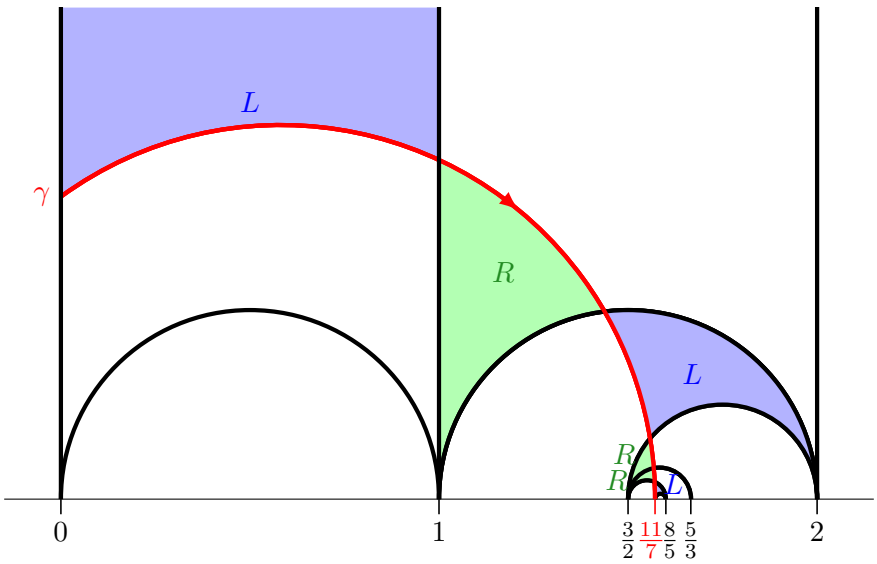

Figure 1: Geodesic algorithm for 11/7

This example illustrates how the continued fraction expansion can be read directly from the interaction of the geodesic in this tessellation.

Moreover, Proposition 2.4 provides the framework to write a Python implementation of the algorithm, encoding the geometry behind it by extracting the exponents of the translations, thus computing the continued fraction. As well as this, the code also gives a LaTeX output for the fraction. This can be found in Appendix A, computing the first n digits of a continued fraction expansion of a given $x \in \mathbb{R}$. Below is the code output when we run it for the example $x = 11/7$.

```
x = 1.5714285714285714
Continued fraction digits: [1; 1, 1, 3]
\[1+\frac{1}{1+\frac{1}{1+\frac{1}{3}}}\]
```

As expected, this yields the same expansion as we calculated previously.

Observe that a geodesic γ generates the same cutting sequence as the geodesic joining ∞ and x after an initial translation into the fundamental region. Hence we can consider the mapping from ∞. This allows us to read the continued fraction from the exponents on T_1 as a map at infinity

$$x = T_1^{n_0} S T_1^{-n_1} S T_1^{n_2} S ... \cdot \infty \Longrightarrow x = [n_0; n_1, n_2, ...]$$

Having established a correspondence between hyperbolic geometry and continued fractions, a natural question to ask is how effective these expansions are in approximating x. This leads us to consider the convergence properties of continued fractions and the quality of their associated approximations.

2.4 Approximating Real Numbers using the Geodesic Algorithm

To discuss convergence and approximation, we first define *convergents* p_n/q_n.

Definition 2.6 (Convergents of a Continued Fraction)**.** *Let $x \in \mathbb{R}$ have continued fraction expansion $x = [a_0; a_1, a_2, ...]$. We define the nth* convergent *of this expansion to be*

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, ..., a_n].$$

Continued fractions are known to yield optimal rational approximations, and this can be quantified through their convergents. In this section, we analyse the convergence behaviour of these approximations and establish bounds on their error.

In the following theorem, we formalise the approximation properties of continued fractions. The convergents of a continued fraction not only converge to the original number, but satisfy strong arithmetic and geometric properties that make them particularly effective approximations. We adapt the proof to explicitly show the result for all n.

Theorem 2.7. *Let $x = [a_0; a_1, a_2, ...]$ and define $p_n/q_n = [a_0; a_1, a_2, ..., a_n]$. Then*

1) $\begin{pmatrix} p_n & p_{n+1} \\ q_n & q_{n+1} \end{pmatrix}$ *has determinant* ± 1 *for all* $n \geq 0$.

2) For $n \geq 1$, $p_{n+1} = a_{n+1}p_n + p_{n-1}$, $q_{n+1} = a_{n+1}q_n + q_{n-1}$ *where* $p_{-1} = 1, q_{-1} = 0$.

3) For all $n \in \mathbb{N}$, *the following inequality holds*

$$p_{2n}/q_{2n} \leq p_{2n+2}/q_{2n+2} \leq x \leq p_{2n+1}/q_{2n+1} \leq p_{2n-1}/q_{2n-1}$$

with equality on one side if and only if x *is rational and the sequence terminates.*

Proof. *1)* We first show that the matrix has determinant ± 1. In particular, this relation tells us that consecutive convergents are Farey neighbours. Let

$$T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

As in [Ser15], define for $n \geq 0$

$$M_n = \begin{cases} T_1^{-a_n} S T_1^{a_{n-1}} \cdots S T_1^{a_1} S T_1^{-a_0}, & \text{if } n \equiv 0 \mod 2 \\ S T_1^{a_n} S T_1^{-a_{n-1}} \cdots S T_1^{a_1} S T_1^{-a_0}, & \text{if } n \equiv 1 \mod 2 \end{cases}$$

That is, for $k \geq 0$, $M_{2k} = T_1^{-a_{2k}} M_{2k-1}$ and $M_{2k+1} = S T_1^{a_{2k+1}} S M_{2k}$. We claim the following.

Claim 2.8. *Let* M_n *be defined as above. Then*

$$M_n[a_0; a_1, a_2, ..., a_n + \xi] = \begin{cases} \xi, & \text{if } n \equiv 0 \mod 2 \\ \frac{1}{\xi} & \text{if } n \equiv 1 \mod 2 \end{cases}$$

Proof. Define $f_n(\xi) = [a_0, a_1, ..., a_n + \xi]$

Notice $S T_1^{a_1} S T_1^{-a_0}[a_0; a_1, a_2, a_3, ...] = S T_1^{a_1}(-[a_1; a_2, a_3, ...]) = [a_2; a_3, ...]$. For n even, let $n = 2k$ and note that

$$S T_1^{a_{2k-1}} S T_1^{a_{2k-2}} ... S T_1^{a_1} S T_1^{a_0} f_n(\xi) = [a_{2k} + \xi] = a_{2k} + \xi$$

Hence, $M_{2k}[a_0; a_1, ..., a_{2k} + \xi] = M_{2k} f_{2k}(\xi) = [\xi] = \xi$. Now suppose that $n =$

$2k+1$. Then

$$\begin{aligned}
M_{2k+1}f_{2k+1}(\xi) &= ST^{a_{2k+1}}SM_{2k}f_{2k+1}(\xi) \\
&= ST_1^{a_{2k+1}}SM_{2k}f_{2k}(1/(a_{2k+1}+\xi)) \\
&= ST_1^{a_{2k+1}}S[1/(a_{2k+1}+\xi)] && \text{(from even case)} \\
&= ST^{a_{2k+1}}(-(a_{2k+1}+\xi)) \\
&= S(-\xi) \\
&= \frac{1}{\xi}
\end{aligned}$$

□

Defining the matrices this way allows us to avoid any problems with signs when evaluating ξ at 0 and ∞. For the even case, it is clear that $M_{2k}[a_0; a_1, ..., a_{2k}] = 0$ and $M_{2k}[a_0; a_1, ..., a_{2k-1}] = \infty$. Hence,

$$\frac{p_{2k}}{q_{2k}} = M_{2k}^{-1}(0), \qquad \frac{p_{2k-1}}{q_{2k-1}} = M_{2k}^{-1}(\infty)$$

Therefore, since M_n is a word in S and T_1, it must have determinant 1. It follows that

$$M_{2k}^{-1} = \begin{pmatrix} p_{2k-1} & p_{2k} \\ q_{2k-1} & q_{2k} \end{pmatrix} \Longrightarrow \det \begin{pmatrix} p_{2k-1} & p_{2k} \\ q_{2k-1} & q_{2k} \end{pmatrix} = 1.$$

Furthermore, $M_{2k+1}[a_0; a_1, ..., a_{2k+1}] = \infty$ and $M_{2k+1}[a_0; a_1, ..., a_{2k}] = 0$. Swapping columns will negate the determinant, hence

$$M_{2k+1}^{-1} = \begin{pmatrix} p_{2k+1} & p_{2k} \\ q_{2k+1} & q_{2k} \end{pmatrix} \Longrightarrow \det \begin{pmatrix} p_{2k} & p_{2k+1} \\ q_{2k} & q_{2k+1} \end{pmatrix} = -1.$$

In particular,

$$p_n q_{n+1} - p_{n+1} q_n = (-1)^{n+1}$$

2) We now establish the recurrence relations. Observe that $[a_0; a_1, a_2, ..., a_n, a_{n+1}] = [a_0; a_1, a_2, ..., a_n + 1/a_{n+1}]$, thus

$$\frac{p_{2k+1}}{q_{2k+1}} = M_{2k}^{-1} \cdot \frac{1}{a_{2k+1}} = \frac{p_{2k-1}\left(\frac{1}{a_{2k+1}}\right) + p_{2k}}{q_{2k-1}\left(\frac{1}{a_{2k+1}}\right) + q_{2k}} = \frac{p_{2k-1} + a_{2k+1}p_{2k}}{q_{2k-1} + a_{2k+1}q_{2k}}.$$

Similarly, noting instead that $M_{2k+1}^{-1}[a_0; a_1, ..., a_{2k+1} + 1/a_{2k+2}] = 1/(1/a_{2k+2}) = a_{2k+2}$,

$$\begin{pmatrix} p_{2k+1} & p_{2k} \\ q_{2k+1} & q_{2k} \end{pmatrix} \cdot a_{2k+2} = \frac{a_{2k+2}p_{2k+1} + p_{2k}}{a_{2k+2}p_{2k+1} + p_{2k}} = \frac{p_{2k+2}}{q_{2k+2}}.$$

From *1)*, we have that p_n and p_{n-1} are coprime as well as q_n and q_{n-1}. We want to prove that the above is in its lowest form, i.e. that $p_{n-1} + a_{n+1}p_n$ and $q_{n-1} + a_{n+1}q_n$ are coprime.

Suppose d is the gcd of the two. Then d must divide any linear combination of the two, in particular,

$$d \mid (q_n(p_{n-1} + a_{n+1}p_n) - p_n(q_{n-1} + a_{n+1}q_n)) \Longrightarrow d \mid (p_{n-1}q_n - p_nq_{n-1})$$

but this is simply the determinant condition, thus d divides 1. Therefore, they must be coprime and the rational expression in its lowest form.

3) Finally, we prove the inequality. Let t be the tail of the continued fraction for x such that $x = [a_0; a_1, ..., a_n + t]$ and $0 \leq t < 1 < \infty$. Since $M_n^{-1} \in PSL_2(\mathbb{Z})$, it is orientation-preserving. Therefore,

$$M_{2k}^{-1}[0, \infty) = \left[\frac{p_{2k}}{q_{2k}}, \frac{p_{2k-1}}{q_{2k-1}}\right), \qquad M_{2k+1}^{-1}[0, \infty) = \left[\frac{p_{2k}}{q_{2k}}, \frac{p_{2k+1}}{q_{2k+1}}\right)$$

and so $M_n^{-1}(0) \leq M_n^{-1}(t) < M_n^{-1}(\infty)$. Thus, $x = M_n^{-1}(t) \in M_n^{-1}[0, \infty)$.

In particular,

$$\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} = \frac{p_nq_{n-1} - p_{n-1}q_n}{q_nq_{n-1}} = \frac{(-1)^{n+1}}{q_nq_{n-1}}$$

Therefore,

$$\frac{p_{2k}}{q_{2k}} \leq \frac{p_{2k+2}}{q_{2k+2}} \leq x \leq \frac{p_{2k+1}}{q_{2k+1}} \leq \frac{p_{2k-1}}{q_{2k-1}}$$

□

The recurrence relations show that the convergents can be computed efficiently, while the determinant condition ensures that successive approximations are in lowest terms. Most importantly, the inequalities demonstrate that the convergents provide increasingly accurate approximations to x, with the error decreasing rapidly as n increases.

As an immediate consequence, we obtain the following bound on the error of the convergents. This corollary is only stated in [Ser15] so we will prove it here.

Corollary 2.9. *With the notation as in Theorem 2.7, $p_n/q_n \to x$ as $n \to \infty$. Moreover, $|x - p_n/q_n| \leq 1/q_nq_{n+1}$, and $q_n \geq n$ for all $n \in \mathbb{N}$.*

Proof. From part *3)* in the above theorem, we can say that

$$\left|x - \frac{p_n}{q_n}\right| \leq \left|\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n}\right| = \frac{|p_{n+1}q_n - p_nq_{n+1}|}{q_nq_{n+1}} = \frac{1}{q_nq_{n+1}}$$

Recall that $q_{n+1} = a_{n+1}q_n + q_{n-1}$. Note that $a_k \geq 1$ for all $k \in \mathbb{N}$, since for some $y \in [0,1)$, $|S \cdot y| > 1$, thus we will always have to translate back into the unit interval after an inversion. Therefore, $q_{n+1} \geq q_n + q_{n-1}$. Since $q_0 = 1$, $q_n \geq F_n$, where F_n is the nth Fibonacci number. The Fibonacci numbers clearly satisfy $F_n \geq n$, thus $q_n \geq n$. Substituting this back into the above expression gives us

$$\left| x - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}} \leq \frac{1}{n^2 + n} \longrightarrow 0$$

□

The error is $O(1/n^2)$ so the error in approximating x by its convergents decreases at least quadratically in the denominator. The algorithm therefore gives very accurate approximations after only a few iterations. Let's consider an example.

Example 2.10. Let $x = \sqrt{5} = [a_0; a_1, a_2, ...]$. Clearly $4 < 5 < 9$ so $2 < \sqrt{5} < 3$, thus $a_0 = 2$.

$$\sqrt{5} = [2; a_1, a_2, ...]$$

$$\sqrt{5} - 2 = [0; a_1, a_2, ...]$$

$$\frac{1}{\sqrt{5} - 2} = \sqrt{5} + 2 = [a_1; a_2, a_3, ...]$$

$4 < \sqrt{5} + 2 < 5$, so

$$\sqrt{5} + 2 = [4; a_2, a_3, ...]$$

$$\sqrt{5} - 2 = [0; a_2, a_3, ...]$$

We can see that $[a_1; a_2, a_3, ...] = [a_2; a_3, a_4, ...]$, so for all $n \geq 1$, $a_n = a_{n+1} = 4$. So

$$\sqrt{5} = [2; 4, 4, 4, ...] = 2 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \cdots}}}$$

We will use the continued fraction to read off approximates for $\sqrt{5}$. Observe that $q_{n+1} = 4q_n + q_{n-1} > 4q_n$, thus we can approximate the error term by

$$\left| \sqrt{5} - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}} \leq \frac{1}{4q_n^2}$$

Recall that we defined $q_{-1} = 0$ and $q_1 = 1$.

- $n = 0$: $|\sqrt{5} - 2| \leq 1/4$. Indeed, $|\sqrt{5} - 2| \approx 0.236$.
- $n = 1$: $|\sqrt{5} - 9/4| \leq 1/4(4)^2 = 1/64$. Indeed, $|\sqrt{5} - 9/4| \approx 0.0139$.

We can see in fact that our error bound is at most $1/4^{2n+1}$, thus the continued fraction provides a very accurate approximation for $\sqrt{5}$ after only a few iterations.

After finding an algorithm to calculate the continued fraction of a real number, the natural next question is to ask if this idea can be adapted for complex numbers. In fact, a very similar geodesic algorithm can be computed for complex numbers. We extend the notion of orientation-preserving isometry to $\mathbb{H}^3$ and use a geodesic γ from ∞ to $\mathbb{C}$. To do so, we first need to find an analogous group and tessellation that allows us to do this.

3 Extending the Möbius Action to Isometries on $\mathbb{H}^3$

To begin extending the geometric continued fraction algorithm to complex numbers, we require an analogue of the hyperbolic plane in higher dimensions, together with a suitable extension of the Möbius action. In this section, we construct an action of $PSL_2(\mathbb{C})$ on hyperbolic three-space $\mathbb{H}^3$ and show that it acts by orientation-preserving isometries. This will provide the geometric framework needed to generalise the Farey tessellation and geodesic algorithm to the complex setting.

Define $\mathbb{H}^3 := \{(z,t) \in \mathbb{C} \times \mathbb{R}^\times\}$. This has boundary $\partial\mathbb{H}^3 = \mathbb{C} \cup \{\infty\} =: \widehat{\mathbb{C}}$. To compute a continued fraction in $\mathbb{C}$, we want to adapt this idea to matrix outputs in the plane $\mathbb{C}$ using an extension of the Möbius action that reduces to the usual one when considering $\mathbb{H}^2$ embedded in $\mathbb{H}^3$ via the two vertical planes $\mathbb{R} \times \mathbb{R}^+$ and $i\mathbb{R} \times \mathbb{R}^+$. We define a quaternion $h = z + tj = x + yi + tj \in \mathbb{H}^3$ where $z = x + iy$. To do so, we need to show that this space is closed as a subset of the quaternions, and then extend the Möbius action.

3.1 Constructing the Extended Möbius Action via Quaternions

$PSL_2(\mathbb{Z})$ acts on the upper-half plane via Möbius transformations. Formally, this is

$$PSL_2(\mathbb{Z}) \curvearrowright \mathbb{H}^2 \quad \text{via} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az+b}{cz+d}.$$

To preserve the properties of the action of $PSL_2(\mathbb{Z})$ on $\mathbb{H}^2$, we want the action of $PSL_2(\mathbb{Z}[i])$ on a quaternion $h = z + tj$ for $z \in \mathbb{C},\ t > 0$ to mirror that of the 2-dimensional case, in particular

$$PSL_2(\mathbb{Z}[i]) \curvearrowright \mathbb{H}^3 \quad \text{via} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot h = \frac{ah+b}{ch+d}.$$

Recall that a common presentation for quaternions is given by

$$\mathcal{Q} \cong \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = -1 \rangle.$$

We will use this fact extensively, in addition to the following two identities that will aid the computation for extending the action to $\mathbb{H}^3$.

Lemma 3.1. *Let $z, w \in \mathbb{C}$, and $\overline{a + bi + cj + dk} = a - bi - cj - dk$ be the quaternionic conjugate. Then we have the following:*

1) $zj = j\overline{z}$

2) $\overline{z + wj} = \overline{z} - wj$

Proof. Note that, from $ijk = -1$, we have that $ij = k$, $i = jk$, and $-i = kj$.

1) It suffices to check that $ij = j\overline{i} = -ji$.

$$ij = (jk)j = j(kj) = j(-i) = -ji$$

2) We now use 1).

$$\begin{aligned}(z + wj)(\overline{z} - wj) &= |z|^2 - zwj + wj\overline{z} - (wj)^2 \\ &= |z|^2 - zwj + wzj - wj(j\overline{w}) \\ &= |z|^2 + |w|^2\end{aligned}$$

Thus, we have $\overline{z + wj} = \overline{z} - wj$

□

We now construct the aforementioned extension of the Möbius action to $\mathbb{H}^3$. While some authors take a more group-theoretic approach, we will instead prove this via a direct computation and explicitly show the action is an orientation-preserving isometry in a more analytic manner. The strategy is to represent points in $\mathbb{H}^3$ using quaternions and define the action in a way that mirrors the classical case in $\mathbb{H}^2$.

Theorem 3.2 (Extended Möbius Action)**.** *Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL_2(\mathbb{C})$. For $(z, t) \in \mathbb{H}^3$, define the action $g \cdot (z, t)$ to be*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot (z, t) = \left(\frac{(az + b)(\overline{cz + d}) + a\overline{c}t^2}{|cz + d|^2 + |c|^2 t^2}, \frac{t}{|cz + d|^2 + |c|^2 t^2} \right) \tag{†}$$

This is a well-defined smooth action of $PSL_2(\mathbb{C})$ on $\mathbb{H}^3$, and each element acts by an isometry.

Proof. First, let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL_2(\mathbb{C})$ that is $ad - bc = 1$. We will represent (z, t) as a quaternion $h = z + tj$. As such, we consider the action $h \mapsto (ah + b)(ch + d)^{-1} = (ah + b)/(ch + d)$. One would expect the extension to be of this form in accordance with the structure of Möbius maps in $\mathbb{H}^2$. We will prove that it has the same desired properties of a Möbius map, namely orientation-preserving isometries, but a dimension higher.

It is important to note that the quaternions are not commutative, unlike the complex numbers, and instead define a division algebra. Division here is defined in the usual way, such that $x/y = xy^{-1}$.

Writing $h = z + tj$, a direct computation using Lemma 3.1 decomposes this into its individual z and j components.

$$\frac{ah+b}{ch+d} = \frac{(az+b)\overline{(cz+d)} + a\bar{c}t^2}{|cz+d|^2 + |c|^2t^2} + \frac{t}{|cz+d|^2 + |c|^2t^2}j$$

This computation can be found in Appendix B since the purpose here is to understand the motivation for choosing this action. The proof that this is a group action is excluded, as it mirrors the proof as in Proposition 2.1.

One can easily check that replacing (a, b, c, d) with $(-a, -b, -c, -d)$ yields the same expression, hence we may now consider g to be in the group $PSL_2(\mathbb{C})$.

We now show that this acts by isometry. Suppose $g \cdot (z, t) = (z', t')$. This action is from $\mathbb{H}^3$ to itself since $|cz + d|^2 + |c|^2t^2 \geq 0$, in particular is equal to 0 if and only if $c = d = 0$. However, $ad - bc = 1$, hence this cannot happen, so $|cz+d|^2+|c|^2t^2 > 0$. Since the formula is the quotient of two smooth functions with positive denominator, the map is smooth.

To show that this is an isometry of $\mathbb{H}^3$, we check that the metric is preserved. Recall that

$$ds^2 = \frac{|dz|^2 + dt^2}{t^2}.$$

Let $f_g : \mathbb{H}^3 \to \mathbb{H}^3$ be the Möbius extension associated to g. Define

$$P := (az+b)(\overline{cz+d}) + a\bar{c}t^2, \qquad Q := |cz+d|^2 + |c|^2t^2$$

such that $z' = P/Q$ and $t' = t/Q$. We aim to show that the Euclidean metric is scaled by a factor of $1/Q^2$ such that

$$\frac{|dz'|^2 + dt'^2}{t'^2} = \frac{(|dz|^2 + dt^2)/Q^2}{t^2/Q^2} = \frac{|dz|^2 + dt^2}{t^2}$$

and thus proving that the hyperbolic metric is preserved. Since the Jacobian of f_g, which we denote as Df_g, provides the best linear approximation to the map at a point, and a Riemannian metric measures angles and lengths, it suffices to examine how the differential acts on inner products. Indeed, for Euclidean vectors v, w, we have $\langle v, w\rangle = v^T w$, hence it follows that

$$\langle Df_g v, Df_g w\rangle = (Df_g v)^T (Df_g w) = v^T((Df_g)^T(Df_g))w. \qquad (*)$$

Thus, $(Df_g)^T(Df_g)$ provides insight into how inner products are transformed under the map, and computing it allows us to determine metric preservation.

To proceed, we will use ideas presented in [Wir27], namely the Wirtinger derivatives. These are given by

$$\frac{\partial}{\partial z} := \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \qquad \frac{\partial}{\partial \bar{z}} := \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

An important result stated in this reference is that the standard differentiation rules apply, and that

$$\frac{\partial}{\partial z}\bar{z} = \frac{\partial}{\partial \bar{z}}z = 0.$$

Therefore, we can treat z and $\bar{z}$ as two independent variables.

Define $w := cz + d$. Note that

$$\partial_z w = c, \qquad \partial_{\bar{z}} = 0, \qquad \partial_z \bar{w} = 0, \qquad \partial_{\bar{z}} \bar{w} = \bar{c}$$

Calculating the partial derivatives with respect to $z, \bar{z}, t$ for Q and P gives us the following:

$$\begin{aligned}
&\partial_z Q = \partial_z(|w|^2) = \partial_z(w\bar{w}) = c\bar{w} && \partial_z P = \partial_z((az+b)\bar{w}) = a\bar{w} \\
&\partial_{\bar{z}} Q = \bar{c}w && \partial_{\bar{z}} P = \bar{c}(az+b) \\
&\partial_t Q = 2|c|^2 t && \partial_t P = 2a\bar{c}t
\end{aligned}$$

We can use the above to calculate the partial derivatives of z' and t' by using the quotient rule and the above partial derivatives on the numerator and denominator. This allows us to calculate the Jacobian of the map.

We can show that

$$aQ - cP = \overline{w}, \qquad (az+b)Q - wP = -\bar{c}t^2$$

the computations of which are provided in Appendix C. Calculating the partial derivatives gives us the following:

$$\begin{aligned}
\frac{\partial z'}{\partial z} &= \frac{Q\cdot\partial_z P - P\cdot\partial_z Q}{Q^2} = \frac{a\bar{w}Q - c\bar{w}P}{Q^2} = \frac{\bar{w}(aQ-cP)}{Q^2} = \frac{\overline{w}^2}{Q^2} \\
\frac{\partial z'}{\partial \bar{z}} &= \frac{Q\cdot\partial_{\bar{z}} P - P\cdot\partial_{\bar{z}} Q}{Q^2} = \frac{\bar{c}(az+b)Q - \bar{c}wP}{Q^2} = \frac{\bar{c}((az+b)Q - wP)}{Q^2} = \frac{-\bar{c}^2t^2}{Q^2} \\
\frac{\partial z'}{\partial t} &= \frac{Q\cdot\partial_t P - P\cdot\partial_t Q}{Q^2} = \frac{2a\bar{c}tQ - 2|c|^2tP}{Q^2} = \frac{2\bar{c}t(aQ-cP)}{Q^2} = \frac{2\bar{c}t\overline{w}}{Q^2}
\end{aligned}$$

$$\frac{\partial t'}{\partial z} = \frac{Q \cdot 0 - t \cdot \partial_z Q}{Q^2} = \frac{-c\bar{w}t}{Q^2}$$
$$\frac{\partial t'}{\partial \bar{z}} = \frac{Q \cdot 0 - t \cdot \partial_{\bar{z}} Q}{Q^2} = \frac{-\bar{c}wt}{Q^2}$$
$$\frac{\partial t'}{\partial t} = \frac{Q \cdot 1 - t \cdot \partial_t Q}{Q^2} = \frac{Q - 2|c|^2 t^2}{Q^2} = \frac{|w|^2 - |c|^2 t^2}{Q^2}$$

Referring back to the definition of the Wirtinger derivatives, it follows that

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}}, \qquad \frac{\partial}{\partial y} = i\left(\frac{\partial}{\partial z} - \frac{\partial}{\partial \bar{z}}\right)$$

A simple computation then yields that the partial derivatives for z' and t' with respect to x and y are given by

$$\frac{\partial z'}{\partial x} = \frac{\bar{w}^2 - \bar{c}^2 t^2}{Q^2} \qquad \frac{\partial t'}{\partial x} = -\frac{(\bar{c}w + c\bar{w})t}{Q^2}$$
$$\frac{\partial z'}{\partial y} = i\left(\frac{\bar{w}^2 + \bar{c}^2 t^2}{Q^2}\right) \qquad \frac{\partial t'}{\partial y} = i\left(\frac{(\bar{c}w - c\bar{w})t}{Q^2}\right)$$

From $*$, calculating $(Df_g)^T(Df_g)$ is equivalent to defining the inner product on the columns of Df_g. Define the columns of Df_g to be ${Df_g}^{(i)}$ for $i \in \{x, y, t\}$. Firstly, note that $\partial_i z'$ is vector with two entries for $i \in \{x, y, t\}$, a real and imaginary part. Letting $z_1 = x_1 + iy_2$ and $z_2 = x_2 + iy_2$, we get

$$\langle (z_1, t_1), (z_2, t_2) \rangle = x_1 x_2 + y_1 y_2 + t_1 t_2 = \mathfrak{Re}(z_1 \overline{z_2}) + t_1 t_2$$

Therefore, looking first at ${Df_g}^{(x)}$ and ${Df_g}^{(y)}$,

$$\begin{aligned}
\langle {Df_g}^{(x)}, {Df_g}^{(y)} \rangle &= \langle (\partial_x z', \partial_x t'), (\partial_y z', \partial_y t') \rangle \\
&= \mathfrak{Re}(\partial_x z' \overline{\partial_y z'}) + \partial_x t' \partial_y t' \\
&= \mathfrak{Re}(-i(\partial_z z' + \partial_{\bar{z}} z')(\overline{\partial_z z'} - \overline{\partial_{\bar{z}} z'})) - \frac{it^2}{Q^4}(\bar{c}w + c\bar{w})(\bar{c}w - c\bar{w}) \\
&= \mathfrak{Im}(\underbrace{|\partial_z z'|^2 - |\partial_{\bar{z}} z'|^2}_{\in \mathbb{R}, \text{ thus vanishes}} + \partial_{\bar{z}} z' \overline{\partial_z z'} - \partial_z z' \overline{\partial_{\bar{z}} z'}) - \frac{it^2}{Q^4}(\bar{c}^2 w^2 - c^2 \bar{w}^2) \\
&= \mathfrak{Im}\left(\frac{-\bar{c}^2 t^2}{Q^2} \cdot \frac{w^2}{Q^2} - \frac{\bar{w}^2}{Q^2} \cdot \frac{-c^2 t^2}{Q^2}\right) - \frac{it^2}{Q^4}(2i\ \mathfrak{Im}(\bar{c}^2 w^2)) \\
&= \frac{t^2}{Q^4}\mathfrak{Im}(-(\bar{c}^2 w^2 - c^2 \bar{w}^2)) + \frac{2t^2}{Q^4}\mathfrak{Im}(\bar{c}^2 w^2) \\
&= \frac{t^2}{Q^4}\mathfrak{Im}(-2i\ \mathfrak{Im}(\bar{c}^2 w^2)) + \frac{2t^2}{Q^4}\mathfrak{Im}(\bar{c}^2 w^2)
\end{aligned}$$

$$
\begin{aligned}
&= -\frac{2t^2}{Q^4}\mathfrak{Im}(\bar{c}^2w^2) + \frac{2t^2}{Q^4}\mathfrak{Im}(\bar{c}^2w^2) \\
&= 0
\end{aligned}
$$

Similarly, one can show that $\langle {Df_g}^{(x)}, {Df_g}^{(t)}\rangle = \langle {Df_g}^{(y)}, {Df_g}^{(t)}\rangle = 0$. The calculation is similar, thus excluded here and can be found in Appendix D.

Lastly, we find the diagonal entries of $(Df_g)^T(Df_g)$. We begin with ${Df_g}^{(x)}$.

$$
\begin{aligned}
|{Df_g}^{(x)}|^2 =& \langle {Df_g}^{(x)}, {Df_g}^{(x)}\rangle \\
&= |\partial_x z'|^2 + (\partial_x t')^2 \\
&= \frac{1}{Q^4}|\bar{w}^2 - \bar{c}^2t^2|^2 + \frac{t^2}{Q^4}(\bar{c}w + c\bar{w})^2 \\
&= \frac{1}{Q^4}\left(|w|^4 + |c|^4t^4 - 2\ \mathfrak{Re}(\bar{w}^2\overline{\bar{c}^2t^2}) + \bar{c}^2w^2t^2 + c^2\bar{w}^2t^2 + 2|c|^2|w|^2t^2\right) \\
&= \frac{1}{Q^4}\left(|w|^4 + |c|^4t^4 - 2t^2\ \mathfrak{Re}(c^2\bar{w}^2) + 2t^2\ \mathfrak{Re}(c^2\bar{w}^2) + 2|c|^2|w|^2t^2\right) \\
&= \frac{1}{Q^4}|(|w|^2 + |c|^2t^2)|^2 \qquad\qquad \text{(Recall that } Q := |w|^2 + |c|^2t^2\text{)} \\
&= \frac{1}{Q^2}
\end{aligned}
$$

A similar computation showing $|{Df_g}^{(y)}|^2 = |{Df_g}^{(t)}|^2 = 1/Q^2$ can be found in Appendix E. Therefore,

$$(Df_g)^T(Df_g) = \frac{1}{Q^2}I_3$$

That is, for vectors $v, w \in \mathbb{H}^3$, $\langle v, w\rangle_{\mathbb{H}^3} = (1/Q^2)\langle v, w\rangle_{\mathbb{R}^3}$, and thus

$$|dz'|^2 + dt'^2 = dx'^2 + dy'^2 + dt'^2 = \frac{1}{Q^2}(dx^2 + dy^2 + dt^2) = \frac{1}{Q^2}(|dz|^2 + dt^2)$$

It follows that

$$\frac{|dz'|^2 + dt'^2}{t'^2} = \frac{(|dz|^2 + dt^2)/Q^2}{t^2/Q^2} = \frac{|dz|^2 + dt^2}{t^2}$$

□

This extension provides the higher-dimensional analogue of the action of $PSL_2(\mathbb{Z})$ on $\mathbb{H}^2$, and forms the foundation for constructing a geometric continued fraction algorithm in the complex setting. In this context, we are concerned with the action on the space rather than the matrices themselves, thus we would like to say that two matrices are the same if their effect on the space is equivalent.

Lemma 3.3. *Let $g, h \in PSL_2(\mathbb{Z}[i])$. Then the action $\dagger$ from $PSL_2(\mathbb{Z}[i])$ to $\mathbb{H}^3$ is injective. That is,*

$$g \cdot (z, t) = h \cdot (z, t)\ \forall (z, t) \in \mathbb{H}^3 \Longrightarrow g = h$$

Proof. It suffices to check that the kernel of the action is trivial. We first check the z component. Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be in the kernel. Then

$$z = \frac{(az+b)\overline{(cz+d)} + a\bar{c}t^2}{|cz+d|^2 + |c|^2 t^2}$$

It follows that $c = 0$ as we have no t component here. Therefore, $|d|^2 z = (az+b)\bar{d}$. Since we have no constant term, $b = 0$, and thus $\forall z \in \mathbb{C}$

$$\begin{aligned} (|d|^2 - a\bar{d})z &= 0 \\ (d-a)\bar{d}z &= 0 \end{aligned}$$

Since $ad - bc = 1$, $\bar{d} \neq 0$, thus $a = d$. Looking at the t-component gives $c = 0$ and $|d|^2 = 1$ providing no additional information, so no further restrictions are required. Therefore, $a^2 = 1$ and the kernel of the action is given by

$$\{g \in PSL_2(\mathbb{Z}[i]) \mid g \cdot (z,t) = (z,t)\ \forall (z,t) \in \mathbb{H}^3\} = \{\pm I\}$$

as required, since $I \sim -I$ in $PSL_2(\mathbb{Z}[i])$. □

From here, we will define f_g to be the map on the space associated to $g \in PSL_2(\mathbb{C})$. That is, $f_g(z,t) = g \cdot (z,t)$. Having established that it preserves the metric and that elements of $PSL_2(\mathbb{Z}[i])$ are entirely defined by their action on the space, we may freely alternate between $g \in PSL_2(\mathbb{C})$ and f_g. We now show that it preserves orientation.

3.2 Orientation-Preservation of the Action

Proving that this action is orientation-preserving completes the analogy with the classical action of $PSL_2(\mathbb{Z})$ on $\mathbb{H}^2$. To determine whether the action preserves orientation, we compute the determinant of the Jacobian of the transformation.

Recall [Lee12], in which we stated that a diffeomorphism between oriented smooth manifolds is orientation-preserving if and only if its Jacobian determinant is everywhere positive. In the case of $\mathbb{H}^3$, the Möbius extensions are diffeomorphisms because they are smooth, bijective mappings from $\mathbb{H}^3$ to itself.

Lemma 3.4. *Let $g \in PSL_2(\mathbb{C})$ and $f_g : \mathbb{H}^3 \to \mathbb{H}^3$ be its extended Möbius action. Then f_g is orientation-preserving.*

Proof. Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL_2(\mathbb{C})$. Let $f_g : \mathbb{H}^3 \to \mathbb{H}^3$ be the extended Möbius map from †, and let $Df_g(q)$ denote the real Jacobian matrix of f_g at $q = (z,t) = (x,y,t) \in \mathbb{H}^3$ with $z = x + iy$.

From the proof of Theorem 3.2 we have

$$(Df_g(q))^T(Df_g(q)) = \frac{1}{Q(q)^2} I_3, \qquad Q(q) = |cz+d|^2 + |c|^2 t^2.$$

Taking determinants and using $\det(A^T A) = \det(A)^2$ gives

$$\det(Df_g(q))^2 = \det((Df_g(q))^T(Df_g(q))) = \det\left(\frac{1}{Q(q)^2} I_3\right) = \left(\frac{1}{Q(q)^2}\right)^3 = \frac{1}{Q(q)^6},$$

hence

$$\det(Df_g(q)) = \pm\frac{1}{Q(q)^3}.$$

In particular, $\det(Df_g(q)) \neq 0$ for all $q \in \mathbb{H}^3$.

Now fix g. The function $q \mapsto \det(Df_g(q))$ is continuous and non-zero on $\mathbb{H}^3$. Since $\mathbb{H}^3$ is also connected, $\det(Df_g(q))$ cannot change sign without vanishing somewhere, thus has constant sign on $\mathbb{H}^3$ for a fixed $g \in PSL_2(\mathbb{C})$.

Using similar reasoning, we now want to show that the sign of the determinant is constant in g for some fixed q. Fix the point $q_0 = (0,1) \in \mathbb{H}^3$ and define the sign of the determinant as a function in g

$$\varphi(g) := \operatorname{sign}\big(\det(Df_g(q_0))\big) \in \{\pm 1\}.$$

The entries of f_g are rational expressions in $(a,b,c,d,z,\bar{z},t)$ with denominator $D(q) > 0$, so the entries of $Df_g(q_0)$ are rational expressions in (a,b,c,d). Hence, $g \mapsto Df_g(q_0)$ and $g \mapsto \det(Df_g(q_0))$ are continuous. Since $\det(Df_g(q_0)) \neq 0$ for all g, it follows that φ is continuous.

We use the fact that $SL_2(\mathbb{C})$ is connected as a topological group without proof, as this is outside the scope of the essay (see [Hil12]). The projection $\pi : SL_2(\mathbb{C}) \to PSL_2(\mathbb{C})$ is a continuous surjection, and hence $PSL_2(\mathbb{C})$ must be connected, since the continuous image of a connected space is also connected. Since $PSL_2(\mathbb{C})$ is connected and φ is continuous, it must be constant because $\{\pm 1\}$ is discrete, so it suffices to check the sign of the determinant for a single g.

Evaluating at the identity element $I_2 \in PSL_2(\mathbb{C})$, we have $f_{I_2} = \mathrm{id}_{\mathbb{H}^3}$, so $Df_{I_2}(q_0) = I_3$ and therefore $\varphi(I_2) = 1$. Hence, $\varphi(g) = 1$ for all $g \in PSL_2(\mathbb{C})$, i.e. $\det(Df_g(q_0)) > 0$ for all g.

We have proved that for each fixed g, the determinant $\det(Df_g(q))$ has constant sign as a function of $q \in \mathbb{H}^3$. But $\det(Df_g(q_0)) > 0$ for every g, therefore $\det(Df_g(q)) > 0$ for all $q \in \mathbb{H}^3$ and all $g \in PSL_2(\mathbb{C})$. By [Lee12], each f_g is orientation-preserving. $\square$

This completes the extension of the classical theory from $\mathbb{H}^2$ to $\mathbb{H}^3$, and ensures that the geometric framework required for our continued fraction algorithm is consistent in the

higher-dimensional setting. Furthermore, the induced boundary map of this action is a Möbius transformation, as expected.

Proposition 3.5. *The induced boundary map $\partial f_g : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ of a Möbius extension is orientation-preserving.*

Proof. Letting $t \to 0$ in † yields, for $cz + d \neq 0$,

$$\partial f_g(z) = \frac{(az+b)\overline{(cz+d)}}{|cz+d|^2} = \frac{az+b}{cz+d},$$

which extends in the usual way to a Möbius transformation of $\widehat{\mathbb{C}}$, sending $-d/c$ to ∞ and ∞ to a/c. Therefore, by Proposition 2.1, the action at the boundary is orientation-preserving. □

Since $PSL_2(\mathbb{Z}[i]) \subset PSL_2(\mathbb{C})$, the results from this chapter hold for $PSL_2(\mathbb{Z}[i])$. In particular, $PSL_2(\mathbb{Z}[i])$ acts on $\mathbb{H}^3$ by orientation-preserving isometries. We now use this action to construct a fundamental domain for $PSL_2(\mathbb{Z}[i])$, which will play a central role in the development of the complex continued fraction algorithm by providing generators and relations for the group.

4 A Fundamental Domain for $PSL_2(\mathbb{Z}[i])$

Recall that to compute continued fractions in $\mathbb{R}$, we considered the Farey tessellation of the upper-half plane model in hyperbolic 2-space built from the elements

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

For us to calculate continued fractions in $\mathbb{C}$, we need to adapt this a dimension higher. The logical extension is to consider $\mathbb{Z}[i]$. With a translation $T_\alpha : z \mapsto z+\alpha$ for $\alpha \in \mathbb{Z}[i]$ and similar inversion as to $S = -1/z$, we expect to be able to generate $PSL_2(\mathbb{Z}[i])$. However, we want to extend the translations from $\mathbb{R}$ to $\mathbb{C}$, thus require the matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad T_i = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$$

to generate $PSL_2(\mathbb{Z}[i])$ as we want a translation in the real direction and in the imaginary direction. We will prove this, and then use a similar idea to Serre [Ser73] but instead of looking at the stabiliser of specific points, we want to look at an equivalent on edges of the polyhedron, which will give us the relations we require on the group.

In this section, we identify a fundamental domain for $PSL_2(\mathbb{Z}[i])$. A fundamental do-

main for the group is a polyhedron generated by the transformations pairing sides of the polyhedron, a notion formalised later in this section. This is demonstrated in Poincaré's Polyhedron Theorem (see [Mas88]), in particular the process identifies a presentation for the group, and thus generators and a full set of relations.

4.1 Description of the Domain

A fundamental domain for $PSL_2(\mathbb{Z})$ is given by the region

$$\left\{ z \in \mathbb{H}^2 : |z| \geq 1, |\mathfrak{Re}(z)| \leq \frac{1}{2} \right\}$$

This can be found in [Ser73]. By analogy with the classical case, we expect a domain bounded by translation strips in the real and imaginary direction, and an inversion sphere. Motivated by this, we construct a polyhedron extending the notion to 3 dimensions. Motivated by this region, we first consider the polyhedron

$$\left\{ (z,t) \in \mathbb{H}^3 : |z|^2 + t^2 \geq 1, |\mathfrak{Re}(z)|, |\mathfrak{Im}(z)| \leq \frac{1}{2} \right\}.$$

However, $PSL_2(\mathbb{Z}[i])$ encodes additional symmetries compared to its two-dimensional analogue. In the classical case, the units of $\mathbb{Z}$ are $\{\pm 1\}$, and hence matrices A and $-A$ in $PSL_2(\mathbb{Z})$ define the same element of $PSL_2(\mathbb{Z})$. In the case of $\mathbb{H}^3$, we may include rotations such as $z \mapsto -z$ via the matrix $R = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$.

As a consequence, the fundamental domain for $PSL_2(\mathbb{Z}[i])$ can be taken to be smaller than one might initially expect, since these additional symmetries allow further identifications of its boundary. Geometrically, allowing for the above rotation gives the polyhedron

$$\mathcal{D} = \left\{ (z,t) \in \mathbb{H}^3 : |z|^2 + t^2 \geq 1, |\mathfrak{Re}(z)| \leq \frac{1}{2},\ 0 \leq \mathfrak{Im}(z) \leq \frac{1}{2} \right\}.$$

The boundary of $\mathcal{D}$ consists of vertical planes corresponding to translations in the real $(\mathcal{R}_+, \mathcal{R}_-)$ and imaginary directions $(\mathcal{I}_0, \mathcal{I}_+)$, together with a spherical face $\mathcal{H}$ arising from inversion.

- $\mathcal{R}_+ = \{(z,t) \in \mathbb{H}^3 : Re(z) = \frac{1}{2},\ 0 \leq Im(z) \leq \frac{1}{2},\ |z|^2 + t^2 \geq 1\}$
- $\mathcal{R}_- = \{(z,t) \in \mathbb{H}^3 : Re(z) = -\frac{1}{2},\ 0 \leq Im(z) \leq \frac{1}{2},\ |z|^2 + t^2 \geq 1\}$
- $\mathcal{I}_+ = \{(z,t) \in \mathbb{H}^3 : -\frac{1}{2} \leq Re(z) \leq \frac{1}{2},\ Im(z) = \frac{1}{2},\ |z|^2 + t^2 \geq 1\}$
- $\mathcal{I}_0 = \{(z,t) \in \mathbb{H}^3 : -\frac{1}{2} \leq Re(z) \leq \frac{1}{2},\ Im(z) = 0,\ |z|^2 + t^2 \geq 1\}$
- $\mathcal{H} = \{(z,t) \in \mathbb{H}^3 : -\frac{1}{2} \leq Re(z) \leq \frac{1}{2},\ 0 \leq Im(z) \leq \frac{1}{2},\ |z|^2 + t^2 = 1\}$

This is illustrated in Figure 2, where the vertical faces and spherical boundary of the

polyhedron are shown.

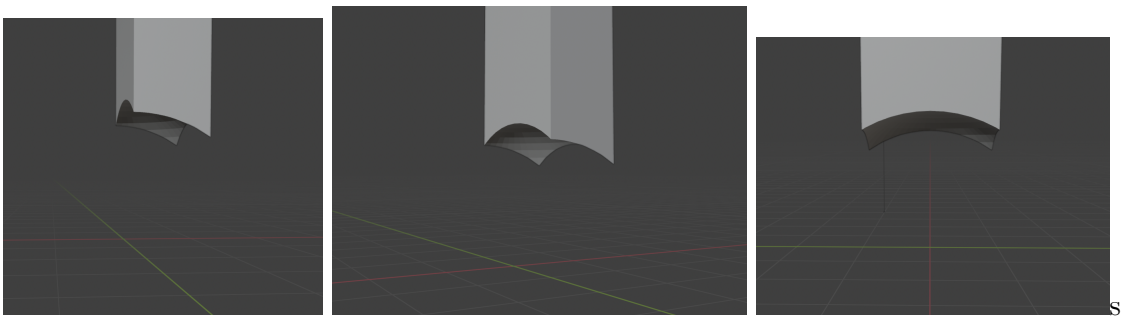

Figure 2: Model of the Polyhedron $\mathcal{D}$

We will prove that $\mathcal{D}$ is in fact a fundamental domain for $PSL_2(\mathbb{Z}[i])$. To do so, we apply Poincaré's Polyhedron Theorem which identifies the group for which the polyhedron is a fundamental domain.

4.2 Geometric Setup for Poincaré's Polyhedron Theorem

We first introduce the geometric ideas describing how the faces of a polyhedron are identified under group elements. To do so, we first need some definitions, taken from [Mas88].

Definition 4.1 (Side Pairing Transformation)**.** *Let D be a polyhedron. Suppose s is a side of a finite-sided polyhedron $D \subset \mathbb{H}^3$. Then there exists a side s', not necessarily distinct from s, and an element $g_s \in Isom(\mathbb{H}^3) = PSL_2(\mathbb{C})$ such that*

(i) $g_s(s) = s'$

(ii) $g_{s'} = g_s^{-1}$

(iii) $g_s(D) \cap D = \varnothing$

Then we call the group element g_s a side pairing transformation.

Intuitively, a side pairing transformation identifies one face of the polyhedron with another via an isometry. Next, we define a cycle transformation as in [Mas88]. Each edge of $\mathcal{D}$ forms a cycle and a similar notion of cycle is defined for infinite tangency points.

Definition 4.2 (Edge Cycle Transformation)**.** *Let $e = e_1$ be some edge of a polyhedron D. Let $e_2 = g_{s_1}(e_1)$ and let s_2 be the side adjacent to s_1' which contains e_2 as an edge. Similarly, let $e_3 = g_{s_2}(e_2)$ and let s_3 be the side adjacent to s_2' which contains e_3 as an edge, and so on. We find that the sequence $e_1, e_2, e_3, ...$ is periodic, with some smallest period k. Then $e_1, e_2, ..., e_k$ is called the edge cycle of e.*
If $g_{s_1}, g_{s_2}, ..., g_{s_k}$ are the corresponding side pairing transformations, then we define the edge cycle transformation of e to be $h := g_{s_k} ... g_{s_1}$.

Definition 4.3 (Infinite Cycle Transformation)**.** *Suppose two sides are tangent at a point on the boundary of $\mathbb{H}^3$. That is, $x_1 \in \partial\mathbb{H}^3 = \mathbb{C} \cup \{\infty\}$. Call one of the sides s_1, let g_1*

be the corresponding side pairing transformation with $g_1(s_1) = s_1'$, and let $x_2 = g_1(x_1)$. If x_2 is a point of tangency of s_1' and some other side $s_2 \neq s_1'$, then let g_2 be the side pairing transformation with $g_2(s_2) = s_2'$, let $x_3 = g_2(x_2)$, and so on. If we eventually have $x_k = x_1$ for some finite k (i.e. $h = g_k ... g_1$ leaves x_1 invariant), then we call x_1 an infinite tangency point and h the infinite cycle transformation.

The side pairing transformations for the polyhedron $\mathcal{D}$ can be identified geometrically. The faces $\mathcal{R}_-$ and $\mathcal{R}_+$ are paired via $g_{\mathcal{R}_-} = T_1$ and $g_{\mathcal{R}_+} = T_{-1} = T_1^{-1}$. The other faces are self-paired via $g_{\mathcal{I}_+} = T_i R, g_{\mathcal{I}_0} = R$, and $g_{\mathcal{H}} = S$.

With these definitions in place, Poincaré's Polyhedron Theorem allows us to determine when such a polyhedron forms a fundamental domain by checking local geometric conditions.

Theorem 4.4 (Poincaré's Polyhedron Theorem (for $\mathbb{H}^3$) [Sie09])**.** *Let $g_s, g_{s'} \in Isom(\mathbb{H}^3) = PSL_2(\mathbb{C})$ and $D \subset \mathbb{H}^3$ a finite-sided polyhedron with side pairing transformations satisfying $(i) - (iii)$ as in the definition. Moreover, the following are satisfied:*

(iv) For each edge e, there is a positive integer t such that the corresponding cycle transformation h satisfies $h^t = 1$ (we call the relations in G of the form $h^t = 1$ the cycle relations).

(v) Let $\alpha(e)$ be the angle measured inside D at the edge e. Then for each edge cycle, we have

$$\sum_{m=1}^{k} \alpha(e_m) = \frac{2\pi}{t}$$

(vi) Every infinite cycle transformation at every infinite tangency point is parabolic.

Then the group G generated by the side pairing transformations is discrete, the cycle relations form a complete set of relations for G, and D is a fundamental domain for G.

In practice, this theorem reduces the problem to verifying that the side pairings are well-defined, that edge cycles have the correct order, and that the associated angles satisfy compatibility conditions.

4.3 Application of Poincaré's Polyhedron Theorem

We now apply Theorem 4.4 to our polyhedron $\mathcal{D}$ to show that it is a fundamental domain for the group G generated by the side pairing transformations. Consequently, we obtain a set of generators and relations for G to aid our construction of a continued fraction algorithm later.

Theorem 4.5 (Fundamental Domain for $PSL_2(\mathbb{Z}[i])$)**.** *The following region $\mathcal{D}$ is a fun-*

damental domain for the group G generated by the side pairing transformations.

$$\mathcal{D} = \left\{ (z,t) \in \mathbb{H}^3 : -\frac{1}{2} \leq Re(z) \leq \frac{1}{2},\ 0 \leq Im(z) \leq \frac{1}{2},\ |z|^2 + t^2 \geq 1 \right\}$$

Proof. The faces $\mathcal{R}_-$ and $\mathcal{R}_+$ are paired via $g_{\mathcal{R}_-} = T_1$ and $g_{\mathcal{R}_+} T_{-1} = T_1^{-1}$ respectively, and the other faces are self-paired via $g_{\mathcal{I}_+} = T_i R$, $g_{\mathcal{I}_0} = R$, $g_{\mathcal{H}} = S$. The self-pairing transformations are self-inverses and not the identity, thus have order 2. Therefore, (i) and (ii) hold from Definition 4.1.

We now verify condition (iii). It suffices to show the interiors are disjoint.

$\mathcal{R}_\pm$: $Im(z)$ and t are unchanged by the element $g_{\mathcal{R}_-} = T_1$ so we only need to consider the real part. The range of $Re(z)$ for $(z,t) \in \mathcal{D}^o$ is $(-1/2, 1/2)$ and, for $T_1(\mathcal{D}^o)$ is $(1/2, 3/2)$. Thus, $\mathcal{D}^o \cap T_1(\mathcal{D}^o) = \varnothing$. A similar argument holds for $g_{\mathcal{R}_+} = T_1^{-1}$.

$\mathcal{I}_+$: $T_i R(z,t) = (-z+i, t)$. The range of $Im(z)$ for $(z,t) \in \mathcal{D}^o$ is $(0, 1/2)$ and for $T_i R(\mathcal{D}^o)$ is $(1/2, 1)$. Thus, $\mathcal{D}^o \cap T_i R(\mathcal{D}^o) = \varnothing$.

$\mathcal{I}_0$: The range of $Im(z)$ for $(z,t) \in \mathcal{D}^o$ is $(0, 1/2)$ and for $(z,t) \in R(\mathcal{D}^o)$ is $(-1/2, 0)$, so again $\mathcal{D}^o \cap R(\mathcal{D}^o) = \varnothing$.

$\mathcal{H}$: Lastly, since $(z,t) \in \mathcal{D}^0$ satisfies $|z|^2 + t^2 > 1$, the intersection is empty if we can show that $|S \cdot (z,t)|^2 < 1$.

$$|S \cdot (z,t)|^2 = \left| \left(\frac{-\bar{z}}{|z|^2 + t^2}, \frac{t}{|z|^2 + t^2} \right) \right| = \frac{|-\bar{z}|^2 + t^2}{(|z|^2 + t^2)^2} = \frac{|z|^2 + t^2}{(|z|^2 + t^2)^2} = \frac{1}{|z|^2 + t^2} < 1$$

So $\mathcal{D}^o \cap S(\mathcal{D}^o) = \varnothing$.

We now verify condition (iv) of Poincaré's Polyhedron Theorem, which requires that the cycle transformations associated to each edge have finite order determined by the dihedral angle at that edge. The edges of $\mathcal{D}$ arise from intersections of pairs of faces, namely intersections of vertical faces, and intersections of vertical faces with the spherical face $\mathcal{H}$.

We begin by computing the dihedral angles between the faces of $\mathcal{D}$. The dihedral angle between any two distinct vertical faces is $\pi/2$. It therefore remains to compute the angles between the spherical face $\mathcal{H}$ and the vertical faces.

Recall that the angle between two surfaces in Euclidean space is given by

$$\cos(\alpha) = \frac{|\langle v_1, v_2 \rangle|}{\|v_1\| \|v_2\|},$$

where v_1, v_2 are normal vectors. Since angles between planes and spheres are preserved between Euclidean and hyperbolic geometry, it suffices to compute these angles in the Euclidean setting.

A normal vector to the unit sphere $\mathcal{H}$ at a point (x, y, t) is given by (x, y, t) itself for $z = x + iy$, while normals to the vertical planes are $(1, 0, 0)$ and $(0, 1, 0)$.

For the edges $\mathcal{H} \cap \mathcal{R}_\pm$ and $\mathcal{H} \cap \mathcal{I}_+$, we obtain $\cos(\alpha) = \frac{1}{2}$, and hence $\alpha = \pi/3$. For $\mathcal{H} \cap \mathcal{I}_0$, we obtain $\cos(\alpha) = 0$, and hence $\alpha = \pi/2$.

We now determine the corresponding cycle transformations. Consider the edge $\mathcal{H} \cap \mathcal{R}_-$. The dihedral angle is $\pi/3$, so six copies of the polyhedron meet around this edge. The associated cycle transformation is given by the composition ST_1, which moves the edge across adjacent faces. Each application advances two copies of the polyhedron; hence three applications return the edge to its original position. Therefore, $(ST_1)^3 = I$. An identical argument applies to the other edges involving $\mathcal{H}$, yielding the relations

$$(ST_1^{-1})^3 = I, \quad (ST_iR)^3 = I, \quad (SR)^2 = I.$$

For edges formed by intersections of vertical planes, the dihedral angle is $\pi/2$, so four copies meet around each edge. The corresponding cycle transformations therefore have order 2, giving the relations

$$(T_1R)^2 = I, \quad (T_1^{-1}R)^2 = I, \quad (T_1T_iR)^2 = I, \quad (T_1^{-1}T_iR)^2 = I.$$

Thus, condition (iv) is satisfied.

(v) follows immediately. Since h is an isometry it preserves angles, and $h^3 = I$,

$$\sum_{k=1}^{2} \alpha(e_m) = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3} = \frac{2\pi}{t} \text{ for edge cycles starting at } \mathcal{H} \cap \mathcal{R}_\pm,\ \mathcal{H} \cap \mathcal{I}_+$$

Here the edge cycle has common period $k = 2$, and traversing this reduced cycle $t = 3$ times yields a full angle 2π, in accordance with condition (v).

Similarly,

$$\sum_{k=1}^{2} \alpha(e_m) = \frac{\pi}{2} + \frac{\pi}{2} = \pi = \frac{2\pi}{2} = \frac{2\pi}{t} \text{ for edges } \mathcal{H} \cap \mathcal{I}_0,\ \mathcal{R}_\pm \cap \mathcal{I}_0,\ \mathcal{R}_\pm \cap \mathcal{I}_+$$

Finally, we show (vi) holds for this polyhedron. That is, every infinite cycle transformation at ∞ is parabolic. The vertical faces are all tangent at $\infty \in \partial\mathbb{H}^3$, and have side pairing transformations T_1, T_1^{-1}, R, and T_iR. These all fix $\infty \in \partial\mathbb{H}^3$. Thus, the stabiliser of ∞ in G is generated by T_1, T_i, and R. We need to consider the infinite cycle transformations at ∞, the only infinite tangency point.

Claim 4.6. *The subgroup of $Stab_G(\infty)$ consisting of parabolic elements is generated by translations T_1 and T_i satisfying $[T_1, T_i] = 1$ (i.e. they commute). Moreover, every infinite cycle transformation lies in this subgroup.*

Proof. A parabolic matrix A is one that satisfies $\mathrm{tr}(A) = 2$, thus translations are parabolic. Commutativity of these translations is trivial.

$$T_1 T_i = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1+i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = T_i T_1$$

Thus, $[T_1, T_i] = I$.

All infinite cycle transformations are products of the side-pairing transformations T_1, R and $T_i R$. These fix the point ∞, meaning they lie in $\mathrm{Stab}(\infty)$, which we have shown is generated by T_1, T_i, and R.

Suppose we have an infinite cycle transformation h starting with a side s_1. Each of T_1 and T_i are translations and thus preserve orientation of the complex plane. We can see that R and $T_i R$ both consist of a rotation π in the complex plane, thus reversing the orientation. Since an infinite cycle must return to the same side with that same side tangent at ∞, the transformation h must preserve the orientation near ∞. Therefore, the number of occurrences of orientation-reversing side pairings must be even, and since they cancel in pairs from the fact that $R^2 = I$ and that $RT_\alpha R = T_{-\alpha} = T_\alpha^{-1}$, the transformation must reduce to a translation, in particular a product of translations T_1 and T_i. Thus we can conclude that h is parabolic.

□

The conditions of Theorem 4.4 hold so we can conclude that the group G associated to this fundamental domain has the following presentation:

$$G \cong \left\langle \; T_1,\, T_i R,\, R,\, S \;\middle|\; \begin{array}{l} (T_i R)^2 = R^2 = S^2 = (ST_1)^3 = (ST_1^{-1})^3 = (SR)^2 = (ST_i R)^3 \\ \qquad\quad = (T_1 R)^2 = (T_1^{-1} R)^2 = (T_1 T_i R)^2 = (T_1^{-1} T_i R)^2 = I, \\ [T_1, T_i] = I \end{array} \right\rangle$$

However, some identities are redundant. We can simplify the presentation, the calculations of which can be found in Appendix F, and the resulting presentation is the following:

$$G \cong \left\langle \; S, T_1, T_i, R \;\middle|\; \begin{array}{l} S^2 = R^2 = (ST_1)^3 = (SR)^2 = (T_1 R)^2 = (T_i R)^2 = (ST_i R)^3 = I, \\ [T_1, T_i] = I \end{array} \right\rangle$$

□

We have established that $\mathcal{D}$ was constructed using the geometric extension of a funda-

mental domain for $PSL_2(\mathbb{Z})$ to provide a fundamental domain for $PSL_2(\mathbb{Z}[i])$. As such, we can now prove that G is a more familiar group, in fact $PSL_2(\mathbb{Z}[i])$.

Proposition 4.7. *With the same notation as in Theorem 4.5, $G = PSL_2(\mathbb{Z}[i])$. In particular,*

$$PSL_2(\mathbb{Z}[i]) \cong \left\langle \; S, T_1, T_i, R \;\middle|\; \begin{array}{l} S^2 = R^2 = (ST_1)^3 = (SR)^2 = (T_1R)^2 = (T_iR)^2 \\ \quad = (ST_iR)^3 = I, \; [T_1, T_i] = I \end{array} \right\rangle$$

is a group presentation.

Proof. For the inclusion $G \subseteq PSL_2(\mathbb{Z}[i])$, note that each generator has determinant 1 and each entry is a Gaussian integer, and thus every element of G must be in $PSL_2(\mathbb{Z}[i])$.

For the other inclusion $PSL_2(\mathbb{Z}[i]) \subseteq G$, let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL_2(\mathbb{Z}[i])$. We aim to find a product U of matrices in G such that $UA = I$ for some $A \in PSL_2(\mathbb{Z}[i])$, then $A = U^{-1}$ and thus is generated by G. We will first consider the case $c \neq 0$, providing a Euclidean algorithm to reduce to a new matrix satisfying bottom-left entry 0. We then follow with the case of $c = 0$.

$(c \neq 0)$ Refer to Algorithm 1 below.

Algorithm 1 Euclidean reduction of the bottom-left entry of a matrix A over $\mathbb{Z}[i]$

Input: $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL_2(\mathbb{Z}[i])$

```
while c ≠ 0 do
    Read (a, b, c, d) from A
    if N(a) < N(c) then
        A ← SA                                  ▷ swap rows: R1 = −R2, R2 = R1
        continue
    end if
    Choose q0, r0 ∈ Z[i] such that c = q0 a + r0 with N(r0) < N(a)
    A ← (S T_{q0} S) A                          ▷ subtract rows: R2 = R2 − q0 R1
    Read (a, b, c, d) from A
    if c = 0 then
        break
    end if
    Choose q1, r1 ∈ Z[i] such that a = q1 c + r1 with N(r1) < N(c)
    A ← T_{−q1} A                               ▷ subtract rows: R1 = R1 − q1 R2
    Read (a, b, c, d) from A
    if a = 0 then
        A ← SA                                  ▷ swap rows: R1 = −R2, R2 = R1
        break
    end if
return A
```

Since $\mathbb{Z}[i]$ is a Euclidean domain, the algorithm will terminate as we are applying

the Euclidean algorithm on each successive remainder. The matrix returned to us will have bottom-left entry 0, so we can now consider the case of $c = 0$.

$(c = 0)$ Since $ad - bc = 1$, $a, d \in \mathbb{Z}[i]$. If $a, d = \pm 1$, then this reduces to the translation $T_{\pm b}$, else after a rotation, for $a = \mp i, d = \pm i$, reduces to a translation $T_{\pm ib}$ since

$$R \begin{pmatrix} -i & b \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & ib \\ 0 & 1 \end{pmatrix}$$

Thus, $PSL_2(\mathbb{Z}[i])$ is generated by S, T_1, T_i, and R. The relations trivially hold as we are considering the same operation in $PSL_2(\mathbb{Z}[i])$ as in G. Thus, $G = PSL_2(\mathbb{Z}[i])$. $\square$

We now have generators for the group. This gives us a way to describe the process of a Gaussian integer continued fraction by translating into some region and then inverting. The next section discusses the tessellation in detail and allows us to see how an inversion behaves in this tessellation.

4.4 Additional Symmetries of the Domain

The fundamental domain constructed above is defined with respect to the action of $PSL_2(\mathbb{Z}[i])$, and the side pairing transformations used in Poincaré's Polyhedron Theorem all lie within this group. However, the geometry admits further symmetries arising from the unit group of the Gaussian integers.

Recall that the units of $\mathbb{Z}[i]$ are given by $\{\pm 1, \pm i\}$. Multiplication by these units induces rotations of the complex plane which preserve the lattice $\mathbb{Z}[i]$. These transformations extend via $\dagger$ to isometries of $\mathbb{H}^3$, and preserve orientation since they are holomorphic.

It is important to note that such rotations are not induced by elements of $PSL_2(\mathbb{Z}[i])$, and therefore do not play a role in the group presentation previously obtained. Nevertheless, they provide additional geometric symmetries that will prove useful for describing a geodesic algorithm in $\mathbb{H}^3$.

In subsequent sections, we will make use of these symmetries to simplify the geometric description by selecting canonical representatives. In particular, we define

$$\kappa = \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix}, \qquad \kappa \cdot z = iz.$$

5 Tessellation of $\mathbb{H}^3$ by Farey Octahedra

Having established a fundamental domain $\mathcal{D}$ for the action of $PSL_2(\mathbb{Z}[i])$ on $\mathbb{H}^3$, we seek a geometric decomposition of hyperbolic three-space analogous to a tessellation of $\mathbb{H}^3$.

Our goal is to extend this correspondence to $\mathbb{H}^3$ by constructing a tessellation using ideal polyhedra adapted to the Gaussian integer setting. In this higher-dimensional context, the role of triangles is played by ideal octahedra, such that the orbit of a single fundamental octahedron $\mathbb{O}_0$ under $PSL_2(\mathbb{Z}[i])$ along with the rotation κ yields a tessellation of $\mathbb{H}^3$, invariant under these actions. This construction provides the geometric framework required to generalise the geodesic algorithm for continued fractions to complex numbers.

5.1 The Fundamental Farey Octahedron $\mathbb{O}_0$

Recall that for $p/q,\ r/s \in \mathbb{Q} \cup \{\infty\} =: \mathbb{Q}_\infty$ reduced to be Farey neighbours, they must satisfy $|ps - rq| = 1$. We extend this notion now to $\mathbb{Q}_\infty(i)$, the set of Gaussian rationals with ∞, defining two reduced Gaussian rationals $\alpha/\beta,\ \delta/\gamma$ to be Farey neighbours if $|\alpha\delta - \beta\gamma| = 1$.

In $\mathbb{H}^2$, the Farey tessellation was given by the orbit of $\mathbb{T}_0 = \{\infty, 0, 1\}$ under $PSL_2(\mathbb{Z})$. The tessellation of $\mathbb{H}^3$ is of octahedra, with the fundamental Farey octahedra defined to have vertex set

$$\mathbb{O}_0 = \left\{\infty, 0, 1, i, 1+i, \frac{1}{1-i}\right\}$$

and can be seen in Figure 3.

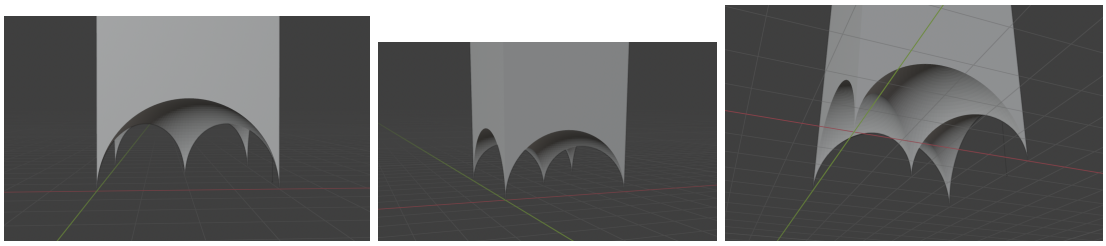

Figure 3: Model of the Fundamental Farey Octahedron $\mathbb{O}_0$

The motivation for this construction is parallel to that of the fundamental domain $\mathcal{D}$ for $PSL_2(\mathbb{Z}[i])$; incorporating the imaginary direction by also considering the triangle $i\mathbb{T}_0 = \{\infty, 0, i\}$, noting that $1/1-i$ is a Farey neighbour with each of $0, 1, i$, and $1+i$.

Definition 5.1 (Farey Sum Set)**.** *Let $\alpha/\beta,\ \gamma/\delta \in \mathbb{Q}_\infty(i)$ be Farey neighbours. We define their Farey sum set to be the set of other possible vertices of a Farey triangle, denoted $\alpha/\beta \oplus_F \gamma/\delta$.*

We know that in $\mathbb{H}^2$, there are two unique Farey triangles whose vertices are pairwise Farey neighbours. That is, there are two elements in the Farey sum set of two Farey neighbours. In $\mathbb{H}^3$, there are instead 4 possible triangles for any two Farey neighbours.

Lemma 5.2. *Let $\alpha/\beta,\ \gamma/\delta \in \mathbb{Q}_\infty(i)$ be Farey neighbours. Then their Farey sum set is given by*

$$\frac{\alpha}{\gamma} \oplus_F \frac{\beta}{\delta} = \left\{\frac{\alpha u + \beta}{\gamma u + \delta} \ :\ u \in \mathbb{Z}[i]^\times\right\}$$

Proof. Consider the vertices $0 = 0/1$ and $\infty = 1/0$. To find the Farey triangles sharing these vertices, we must find the other vertices of the form $\alpha/\beta \in \mathbb{Q}_\infty(i)$ such that the Farey neighbour condition is satisfied. That is, $|0 \cdot \beta + 1 \cdot \alpha| = |\alpha| = 1$ and $|\alpha \cdot 0 + \beta \cdot 1| = |\beta| = 1$. Clearly this tells us that $0 \oplus_F \infty = \mathbb{Z}[i]^\times$. Since the action of $\mathbb{H}^3$ acts by isometries, for an action $g \in PSL_2(\mathbb{Z}[i])$, $|g \cdot (0 \oplus_F \infty)| = 4$.

It is easy to check that

$$\begin{pmatrix} \beta & -\alpha \\ \delta & -\gamma \end{pmatrix} \cdot \left\{ \frac{\alpha}{\beta}, \frac{\gamma}{\delta} \right\} \mapsto \{0, \infty\}$$

Thus defining $M = \begin{pmatrix} \beta & -\alpha \\ \delta & -\gamma \end{pmatrix}$, the set $M^{-1}(\mathbb{Z}[i]^\times)$ identifies the 4 unique vertices of a Farey triangle sharing α/β and γ/δ as vertices.

$$\begin{aligned} M^{-1} \cdot (0 \oplus_F \infty) &= \begin{pmatrix} -\gamma & \alpha \\ -\delta & \beta \end{pmatrix} \cdot \{1, -1, i, -1\} \\ &= \left\{ \frac{-\alpha + \gamma}{-\beta + \delta}, \frac{\alpha + \gamma}{\beta + \delta}, \frac{\alpha i + \gamma}{\beta i + \delta}, \frac{-\alpha i + \gamma}{-\beta i + \delta} \right\} \end{aligned}$$

Thus the Farey sum set is

$$\frac{\alpha}{\beta} \oplus_F \frac{\gamma}{\delta} = \left\{ \frac{\alpha u + \gamma}{\beta u + \delta} \ : \ u \in \mathbb{Z}[i]^\times \right\}$$

□

5.2 Generation of the Tessellation

Rather than constructing the tessellation of $\mathbb{H}^3$ explicitly, we appeal to the work of Hockman [Hoc19], who shows that this space admits a tessellation by octahedra arising naturally from the action of $PSL_2(\mathbb{Z}[i])$ and κ. In particular, the orbit of $\mathbb{O}_0$ under these elements produces the Farey tessellation of $\mathbb{H}^3$. Elements of this orbit are called *Farey octahedra*. The Farey tessellation of $\mathbb{H}^3$ is formalised in the following theorem.

Theorem 5.3 ([Hoc19]). *$\mathbb{H}^3$ is tessellated by Farey octahedrons. That is, any two Farey octahedrons are either equal or disjoint and the union of the closure of the Farey octahedrons covers the whole of $\mathbb{H}^3$.*

This tessellation may be viewed as a three-dimensional analogue of the Farey tessellation of $\mathbb{H}^2$. In particular, the vertices of the octahedra lie on the boundary $\partial\mathbb{H}^3 = \widehat{\mathbb{C}}$ and correspond to elements of $\mathbb{Q}_\infty(i)$. As in the two-dimensional case, this will play a central role in the construction of a continued fraction algorithm for complex numbers.

In particular, the above theorem, together with the construction of the tessellation as the

orbit of $\mathbb{O}_0$ under $PSL_2(\mathbb{Z}[i])$ and κ, gives invariance under $PSL_2(\mathbb{Z}[i])$ and κ. Since the continued fraction algorithm is defined by tracking how geodesics intersect the tessellation, it is crucial that this structure is preserved under the action of $PSL_2(\mathbb{Z}[i])$ and κ. This ensures that the tracking faces, and thus the resulting continued fraction expansion, is well-defined and does not depend on the particular choice of representative or position in $\mathbb{H}^3$.

In the following section, we make this correspondence explicit by analysing how geodesics in $\mathbb{H}^3$ intersect the octahedral tessellation, allowing us to extract continued fraction expansions in $\mathbb{C}$.

6 Gaussian Integer Continued Fractions (GICF)

For a sequence α_n of Gaussian integers, we aim to admit a continued fraction for $z \in \mathbb{C}$ of the form

$$z = \alpha_0 + \frac{1}{\alpha_1 + \frac{1}{\alpha_2 + \dots}}$$

We now have the result that $\mathbb{H}^3$ is tessellated by Farey octahedra. The purpose of this section is to examine this tessellation to develop an algorithm for calculating the continued fraction of a complex number by tracking the faces of the octahedra crossed by the geodesic from ∞ to z. While [Hoc19] begins to do so, this section will aim to develop Hockman's result further and cover full generality, in particular how an appearance of κ alters the continued fraction. Using the generalised algorithm, our aim is to develop Python code reflecting the geometric approach of our algorithm, a result analogous to Proposition 2.4.

We again use the generators of the group to do so. However, one can show that

$$\phi = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = T_i S T_i^{-1} S T_i$$

such that $\phi \cdot z = 1/z$ which represents an action of $PSL_2(\mathbb{Z}[i])$, and it preserves orientation of the tessellation by Corollary 3.5. This allows us to avoid having to alternate signs of each geometric step as we did for the continued fraction of real numbers.

We aim for a similar procedure of the classical continued fraction, reducing to a fundamental unit square and applying an inversion. However, doing so directly loses the structure of the Farey tessellation, as there are 8 Farey triangles in the fundamental octahedron $\mathbb{O}_0$, thus it does not keep track of what triangles we cross. We instead consider an approach similar to [Hoc19]. In contrast, this reflects the geometric structure of the Farey tessellation.

6.1 Naïve Approach to an Algorithm in $\mathbb{H}^3$

We have shown in Theorem 4.5 that $PSL_2(\mathbb{Z}[i])$ can be generated by S, T_1, T_i, and R. The naïve analogous approach to find a continued fraction would be

1) Translation in the real direction

2) Translation in the imaginary direction

3) Single inversion

Here, we would consider the standard Farey triangle tessellation of $\mathbb{H}^2$ as a restriction of $\mathbb{H}^3$. However, there are more considerations we must account for, such as choosing the "fundamental region" that we are mapping into. Using intuition from $\mathbb{H}^2$, we would map into the positive or negative unit square depending on the sign of the real or imaginary part, in turn neglecting the geometric viewpoint with the tessellation of $\mathbb{H}^3$. However, this would in fact not be a Eucliden algorithm as demonstrated in the following example.

Example 6.1. Let $z = \dfrac{(1+\sqrt{2})(1+i)}{2\sqrt{2}} \in \mathbb{C}$. Both $|\mathfrak{Re}(z)| < 1$ and $|\mathfrak{Im}(z)| < 1$, so no translation is needed and we instead invert first.

$$S \cdot z = -\frac{\sqrt{2}(1-i)}{1+\sqrt{2}}$$

Observe that $|\mathfrak{Re}(S \cdot z)|, |\mathfrak{Im}(S \cdot z)| < 1$. Note that the modulus is given by

$$|z| = \frac{1+\sqrt{2}}{2} > 1.$$

Thus before an inversion, we have a remainder greater than 1, so this is not in fact a Euclidean algorithm. Therefore, we would not as easily be able to prove the algorithm terminates. Choosing an alternative fundamental region may resolve this; however the geometry is still ignored in this case.

In fact, a similar method is often used, known as the Hurwitz algorithm as seen in [Osw14]. Here the fundamental region is chosen to be $\{z \in \mathbb{C} : -1/2 \leq \mathfrak{Re}(z), \mathfrak{Im}(z) < 1/2\}$, thus having a similar essence to the real case, but a different domain chosen.

We instead look towards [Hoc19] for inspiration on a geometric algorithm. This perspective allows us to prove termination and results about convergents.

6.2 Relationship between $\mathbb{T}_0$ and $\mathbb{O}_0$

We now want to understand a suitable way of inverting some $z \in \mathbb{C}$ after a translation into the boundary of $\mathbb{O}_0$ that allows us to track the triangles crossed. Dividing this square with vertices $\{0, 1, i, 1+i\}$ into distinct Farey triangles simplifies the problem as we can then look at how each of the triangles maps from $\mathbb{T}_0$. This allows to work in a more familiar

environment as in Section 2.

For the 4 non-vertical faces of $\mathbb{O}_0$, we proceed similarly to Hockman, but we define them to be ordered anticlockwise around the complex square. This allows for improved readability. We define the elements

$$f_1 = \begin{pmatrix} 1 & 0 \\ 1-i & i \end{pmatrix}, \; f_2 = \begin{pmatrix} 1 & 0 \\ -i & 1 \end{pmatrix}, \; f_3 = \begin{pmatrix} 1 & -1+i \\ 1-i & i \end{pmatrix}, \; f_4 = \begin{pmatrix} 1-i & i \\ -i & 1+i \end{pmatrix}.$$

As an action on $\mathbb{H}^3$, these are represented by

$$f_1 = \phi T_{1-i} \phi \kappa^3, \qquad f_2 = \phi T_{-i} \phi, \qquad f_3 = \phi T_{1-i} \phi T_{-1-i} \kappa, \qquad f_4 = T_{1+i} \phi T_{-1+i}$$

We note that the formulation as in [Hoc19] leads to inconsistencies in the algorithm, namely κ in f_1 (defined as f_3 by Hockman). We therefore use the above definition, writing κ^3.

Acting on $\mathbb{T}_0$ gives the other faces of $\mathbb{O}_0$ by identifying their vertex sets. Indeed,

$$f_1\mathbb{T}_0 = \left\{ \frac{1}{1-i}, 0, 1 \right\} \qquad f_3\mathbb{T}_0 = \left\{ \frac{1}{1-i}, 1+i, i \right\}$$

$$f_2\mathbb{T}_0 = \left\{ i, 0, \frac{1}{1-i} \right\} \qquad f_4\mathbb{T}_0 = \left\{ 1+i, \frac{1}{1-i}, 1 \right\}$$

These are the 4 subregions contained in $\mathcal{R} := \{z \in \mathbb{C} : 0 \leq \mathfrak{Re}(z), \mathfrak{Im}(z) < 1\}$. As a subset of the complex plane, this can be observed in Figure 4 below.

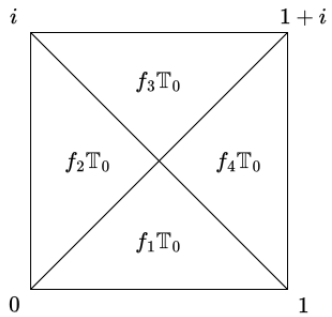

Figure 4: Subregions of $\mathcal{R}$ divided by face maps f_i

In addition to these maps, we will consider $f_5 = \phi$, the map reorientating $\mathbb{T}_0$ in the unique instance of a translation to 0.

Before introducing the algorithm, we first show that the vertices of the Farey tessellation correspond to the Gaussian rationals $\mathbb{Q}(i)$. This identification is crucial, as it provides a notion of convergents in the complex setting, which in turn enables the construction of a continued fraction algorithm.

Proposition 6.2. *$z \in \partial\mathbb{H}^3$ is a vertex of the Farey tessellation if and only if $z \in \mathbb{Q}_\infty(i)$. That is, $\mathbb{Q}_\infty(i)$ is the set of points on the boundary of $\mathbb{H}^3$ such that they appear as vertices of a Farey triangle.*

Proof. The tessellation is obtained by applying elements of $PSL_2(\mathbb{Z}[i])$ to the fundamental triangle $\mathbb{T}_0 = \{\infty, 0, 1\}$. Each element of $PSL_2(\mathbb{Z}[i])$ acts on $\widehat{\mathbb{C}}$ by a Möbius transformation

$$M(z) = \frac{az+b}{cz+d} \qquad (a, b, c, d \in \mathbb{Z}[i]).$$

If $z = \dfrac{\alpha}{\beta} \in \mathbb{Q}_\infty(i)$ then

$$M(z) = \frac{a\alpha + b\beta}{c\alpha + d\beta}$$

for some $\alpha, \beta \in \mathbb{Z}[i]$, so $M(z) \in \mathbb{Q}_\infty(i)$. Since the vertices of $\mathbb{T}_0$ are $\infty, 0, 1 \in \mathbb{Q}_\infty(i)$, it follows that the vertices of every image $M(T_0)$ are Gaussian rationals.

For the other direction, suppose we have $z = \alpha/\beta \in \mathbb{Q}_\infty(i)$. $\mathbb{Z}[i]$ is a Euclidean domain, so there exists some $a, b \in \mathbb{Z}[i]$ such that $a\alpha - b\beta = 1$. Since we know that ∞ is a vertex, then defining $M = \begin{pmatrix} \alpha & a \\ \beta & b \end{pmatrix} \in PSL_2(\mathbb{Z}[i])$ gives us

$$M \cdot \infty = \frac{\alpha}{\beta} = z$$

as required. □

We can now look at the geodesic algorithm. An important distinction is that, unlike in $\mathbb{R}$, we do not have a notion of "left" and "right" in $\mathbb{C}$. Therefore, the algorithm is more difficult to give a direct description of, and is instead easier to consider case-by-case, since we must choose a specific map determined on where in $\mathbb{O}_0$ we translate. As such, we must interpret the continued fraction alternatively to reading a "cutting sequence".

6.3 The Geodesic Algorithm for GICFs

We now describe the geodesic algorithm associated to the Farey octahedral tessellation of $\mathbb{H}^3$. The algorithm repeatedly normalises the endpoint $z \in \mathbb{C}$ by translation into the region $\mathcal{R}$ and applies one of the face maps $\{f_1, f_2, f_3, f_4, f_5\}$. Unlike the 2-dimensional case, we are not applying a universal inversion map after translating; the inversion depends upon which subregion of $\mathcal{R}$ we translate into.

Instead, we will provide motivation for choosing the algorithm, and prove that it is well-defined and that the convergents do in fact converge to $z \in \mathbb{C}$.

Suppose $z \in \mathbb{C}$. Let γ be the geodesic joining ∞ to $z \in \mathbb{C}$. Note that γ begins in the octahedron $T_{a_0}\mathbb{O}_0$ for some $a_0 = \lfloor \mathfrak{Re}(\xi) \rfloor + \lfloor \mathfrak{Im}(\xi) \rfloor \in \mathbb{Z}[i]$. Mapping into $\mathcal{R}$ by $T_{-a_0} = T_{a_0}^{-1}$

allows us to work with $\mathbb{O}_0$. From our definition of the f_i, we see that $f_i^{-1}T_{a_0}^{-1}(z) \in \mathbb{T}_0$ for some f_i. Repeating this process records each face of an octahedron that is crossed, constructing the continued fraction. We now describe this process in Algorithm 6.3.

Algorithm 2 Geodesic Algorithm for Gaussian Continued Fractions

Input: $z \in \widehat{\mathbb{C}}$
Output: A sequence of transformations (g_n) such that $g_n ... g_1 \cdot z = \infty$

```
Initialise an empty list L
z_0 ← z
n ← 0
while z_n ≠ ∞ do
    Translate into the fundamental domain
    Choose α_n ∈ Z[i] such that z_n − α_n ∈ R
    Append T_{−α_n} to L
    if z_n − α_n = 0 then
        break
    end if
    Let f_n ∈ {f_1, f_2, f_3, f_4, f_5} be the transformation corresponding to the region of R
containing z_n − α_n
    Append f_n to L
    z_{n+1} ← f_n^{−1}(z_n − α_n)
    n ← n + 1
end whilereturn L
```

Multiplying the entries of $\mathcal{L}$ gives the product of matrices that map ∞ to z. Importantly, since T_α and κ are invariant at ∞, the first map must be ϕ for this algorithm to be well-defined.

Remark 6.3. *If $T_{-\alpha_n}(z_n)$ lies on the boundary between two subregions, the choice of g_n may not be unique. One may impose a tie-breaking convention, or note that this can occur only on a boundary set of measure zero.*

Lemma 6.4. *Let $z \in \mathbb{C}$. The geodesic algorithm for z terminates if and only if $z \in \mathbb{Q}(i)$.*

Proof. Suppose that the algorithm terminates. That is, there exist some $M \in PSL_2(\mathbb{Z}[i])$ that is a finite word of T_1, T_i, and f_i for $i \in \{1, 2, 3, 4\}$ such that $M(\infty) = z$. Then M is of the form

$$M = \begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{Z}[i]$$

In particular, $z = M(\infty) = \alpha/\beta \in \mathbb{Q}(i)$.

Now suppose $z \in \mathbb{Q}(i)$. Then z is the vertex of some Farey triangle in the tessellation of $\mathbb{H}^3$ by Proposition 6.2. The geodesic will therefore cross finitely many faces before reaching this triangle, terminating the algorithm. □

This result is the Gaussian analogue of the classical fact that an ordinary continued fraction

expansion terminates precisely when the endpoint is rational. Let's look at an example.

Example 6.5. Let $z = z_0 = \dfrac{2-5i}{2+3i} = \dfrac{(2-5i)(2-3i)}{26} = -\dfrac{11}{13} - \dfrac{16}{13}i$. The process of the algorithm is as follows:

1) $z_1 = (T_{-1-2i})^{-1}(z_0) = \dfrac{2}{13} + \dfrac{10}{13}i \in \mathcal{R}$. Observe that, using Figure 4, we apply f_2^{-1}. Thus, $z_1 = f_2^{-1}(T_{-1-2i}^{-1})(z_0) = 2+2i$

2) $(T_{2+2i})^{-1}(z_1) \in \mathcal{R}$, so again by observing where in $\mathcal{R}$ it is mapped to, we then map with $f_5 = \phi$ such that $\infty = z_2 = (T_{2+2i})^{-1} f_2^{-1}(T_{-1-2i}^{-1})(z_0)$

We therefore have that $z = T_{-1-2i} f_2 T_{2+2i}(\infty) = T_{-1-2i}\phi T_{-i}\phi T_{2+2i}\phi(\infty)$, and thus the continued fraction is

$$\frac{2-5i}{2+3i} = [-1-2i; -i, 2+2i] = -1-2i + \frac{1}{-i + \frac{1}{2+2i}}$$

Indeed, the final transform is

$$T_{-1-2i}\phi T_{-i}\phi T_{2+2i}\phi \cdot \infty = \begin{pmatrix} -2+5i & 1+i \\ -2-3i & -1 \end{pmatrix} \cdot \infty = \frac{2-5i}{2+3i}$$

We can see in this example that we had no instance of $\kappa \cdot z = iz$. The addition of this rotation changes how we read the continued fraction, evident in the following lemma.

Lemma 6.6. *Let $T_\alpha \cdot z = z + \alpha,\ \phi \cdot z = 1/z,\ \kappa \cdot z = iz$. Then*

$$\kappa(T_{\alpha_0}\phi T_{\alpha_1}\phi ... \phi T_{\alpha_n}\phi) = T_{i\alpha_0}\phi T_{-i\alpha_1}\phi ... \phi T_{i^{2n+1}\alpha_n}\phi\kappa$$

Consequently,

$$\kappa^r \prod_{m=0}^{n} (T_{\alpha_m}\phi) = \prod_{m=0}^{n} \left(T_{i^{(2m+1)r}\alpha_m}\phi\right)\kappa^r$$

Proof. We will induct on r. We will use the trivial identities $\kappa^r T_\alpha \kappa^{-r} = T_{i^r\alpha}$ and $\kappa\phi = \kappa^2\phi\kappa$. Observing that $\kappa^2\phi = \phi\kappa^2$, we see that, for the base case,

$$\begin{aligned}
\kappa(T_{\alpha_0}\phi T_{\alpha_1}\phi T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi) &= (\kappa T_{\alpha_0}\kappa^{-1})(\kappa\phi)T_{\alpha_1}\kappa^{-1}\kappa\phi T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi \\
&= T_{i\alpha_0}(\kappa^2\phi\kappa)T_{\alpha_1}\kappa^{-1}\kappa\phi T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi \\
&= T_{i\alpha_0}(\kappa^2\phi)(\kappa T_{\alpha_1}\kappa^{-1})\kappa\phi T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi \\
&= T_{i\alpha_0}\phi\kappa^2 T_{i\alpha_1}\kappa\phi T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi \\
&= T_{i\alpha_0}\phi\kappa^2 T_{i\alpha_1}(\kappa^{-2}\kappa^2)\kappa\phi T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi \\
&= T_{i\alpha_0}\phi T_{-i\alpha_1}\kappa^2(\kappa\phi)T_{\alpha_2}\phi ... \phi T_{\alpha_n}\phi
\end{aligned}$$

$$= T_{i\alpha_0}\phi T_{-i\alpha_1}\phi\kappa^4\phi\kappa T_{\alpha_2}\phi...\phi T_{\alpha_n}\phi$$
$$= T_{i\alpha_0}\phi T_{-i\alpha_1}\phi\kappa(T_{\alpha_2}\phi...\phi T_{\alpha_n}\phi) \qquad \text{(Note } \kappa^4 = \phi^2 = I\text{)}$$

at which point the same argument applies to the word $T_{\alpha_2}\phi...\phi T_{\alpha_n}\phi$. It follows that:

$$\text{Base Case:} \qquad \kappa\prod_{m=0}^{n}(T_{\alpha_m}\phi) = \prod_{m=0}^{n}\left(T_{(-1)^m i\alpha_m}\phi\right) = \prod_{m=0}^{n}(T_{i^{2m+1}\alpha_m}\phi)\kappa$$
$$\text{Inductive Assumption:} \qquad \kappa^r\prod_{m=0}^{n}(T_{\alpha_m}\phi) = \prod_{m=0}^{n}(T_{i^{(2m+1)r}\alpha_m}\phi)\kappa^r$$

Therefore, we have that

$$\begin{aligned}
\kappa^{r+1}\prod_{m=0}^{n}(T_{\alpha_m}\phi) &= \kappa\left(\kappa^r\prod_{m=0}^{n}(T_{\alpha_m}\phi)\right)\\
&= \kappa\left(\prod_{m=0}^{n}T_{i^{(2m+1)r}\alpha_m}\phi\right)\kappa^r\\
&= \left(\prod_{m=0}^{n}T_{(i^{2m+1})i^{(2m+1)r}\alpha_m}\phi\right)\kappa^{r+1}\\
&= \left(\prod_{m=0}^{n}T_{i^{(2m+1)(r+1)}\alpha_m}\phi\right)\kappa^{r+1}
\end{aligned}$$

□

We know that the role of ϕ is analogous to the inversion S in the real case, with the additional benefit that we do not have to track alternating signs of the translations. Decomposing the f_i into its individual components as well as the above lemma allows us to read the continued fraction as in the real case using a reduced word in ϕ, T_1, and T_i.

Theorem 6.7. *Let $z \in \mathbb{C}$. Suppose Algorithm 6.3 produces a word*

$$W = g_1g_2g_3...$$

expressed in T_1, T_i, ϕ, and κ such that $W \cdot \infty = z$. Then we can reduce this word to some W' such that

$$W' = T_{\alpha_0}\phi T_{\alpha_1}\phi T_{\alpha_2}\phi...$$

is a word in only T_1, T_i, and ϕ such that $W' \cdot \infty = z$. In particular, the occurrences of ϕ separate successive translations such that $z = [\alpha_0; \alpha_1, \alpha_2, ...]$.. and the final element of a finite word is ϕ.

Proof. By Lemma 6.6, any word W can be written as $W_0\kappa^r$, where W_0 is a word in T_1, T_i, and ϕ, and $r \in \{0, 1, 2, 3\}$. Since $\kappa \in \text{Stab}(\infty)$, the factor κ^r does not affect the value of

$W \cdot \infty$, nor the continued fraction of z. Thus, we only need to consider W_0.

Clearly consecutive translations can be combined into a single translation, so it suffices to check that we proceed to the next entry of a continued fraction when we cross a single ϕ in the word. Since ϕ has order 2, we have that

$$T_\alpha \phi \phi T_\beta = T_\alpha T_\beta = T_{\alpha+\beta}.$$

Hence, any occurrence of consecutive ϕ may be removed, and adjacent translations can be combined. Repeatedly applying these reductions gives us the word

$$W_0' = T_{\alpha_0} \phi T_{\alpha_1} \phi T_{\alpha_2} \phi ...$$

where $\alpha_k \in \mathbb{Z}[i]$ and no two ϕ occur consecutively. By construction of the algorithm, each occurrence of ϕ corresponds to an inversion step in the geodesic algorithm, separating successive translations. Hence the continued fraction expansion is given by

$$z = [\alpha_0; \alpha_1, \alpha_2, ...]$$

□

This mirrors the classical case from Proposition 2.4 where inversions separate translations and encode the digits of a continued fraction. We now consider a more illustrative example.

Example 6.8. Let $z = z_0 = \dfrac{2+3i}{7-2i} = \dfrac{8}{53} + \dfrac{25}{53}i$. We implement the algorithm

1) Observe that $z \in \mathcal{R}$, in particular $z \in f_2 \mathbb{T}_0$. Set $z_1 = f_2^{-1} z_0$

2) $z_1 = \dfrac{1}{2} + \dfrac{3}{4}i \in f_3 \mathbb{T}_0 \in \mathcal{R}$, so set $z_2 = f_3^{-1} z_1$

3) $z_2 = \dfrac{1}{2} + \dfrac{3}{2}i$. First translating into $\mathcal{R}$ with T_i^{-1}, we can apply f_3 to the result, at which point the algorithm terminates since $f_3\left(\dfrac{1}{2} + \dfrac{1}{2}i\right) = \infty$. Thus, $\infty = z_3 = f_3^{-1} z_2$

Therefore, we have that

$$\begin{aligned}
z &= f_2 f_3 T_i f_3(\infty) \\
&= (\phi T_{-i} \phi)(\phi T_{1-i} \phi T_{-1-i} \kappa)(T_i)(\phi T_{1-i} \phi T_{-1-i} \kappa)(\infty) \\
&= \phi T_{-i} T_{1-i} \phi T_{-1-i} T_{i \cdot i} \phi T_{-i \cdot (1-i)} \phi T_{i \cdot (-1-i)} \kappa^2(\infty) \\
&= \phi T_{1-2i} \phi T_{-1-i} T_{-1} \phi T_{-1-i} \phi T_{1-i}(\infty) \\
&= \phi T_{1-2i} \phi T_{-2-i} \phi T_{-1-i} \phi(\infty)
\end{aligned}$$

Since our expansion begins with ϕ, this is equivalent to having $T_0 \phi ...$, i.e. the first entry

in the expansion is 0. Hence the Gaussian integer continued fraction of z is

$$\frac{2+3i}{7-2i} = [0; 1-2i, -2-i, -1-i] = \frac{1}{1-2i+\frac{1}{-2-i+\frac{1}{-1-i}}}$$

We can implement Algorithm 6.3 as Python code. The full implementation is excluded due to length constraints. Instead, the complete code is made publicly available in this GitHub repo.
The code follows the geodesic algorithm, preserving the algebraic and geometric properties. In particular, the output displays the transformation of this geodesic mapping ∞ to z as a word in $T_1, T_i, \phi, \kappa, f_1, f_2, f_3, f_4$. In addition to this, it reduces this word to one alternating between ϕ and a translation using $\phi^2 = I$ and Lemma 6.6, and that $T_\alpha, \kappa \in \mathrm{Stab}(\infty)$.

We explore another example to highlight the code in action. It is natural to ask if this geodesic algorithm in $\mathbb{H}^3$ reduces to the classical algorithm in $\mathbb{H}^2$ when $z \in \mathbb{R}$, so we look at implementing the code for Example 2.5.

Example 6.9. Let $z = 11/7$. Having previously found a continued fraction expansion for this using the classical algorithm, we now look at applying the complex algorithm.

```
--- SYMBOLIC WORD ---
T(1) f1 f4 f2 f2 f3

--- DECOMPOSED WORD ---
T(1) phi T(1-1i) phi kappa kappa kappa T(1+1i) phi T(-1+1i) phi T
    (-1i) phi phi T(-1i) phi phi T(1-1i) phi T(-1-1i) kappa

--- REDUCED WORD ---
T(1) phi T(1-1i) phi T(1-1i) phi T(-1-1i) phi T(-3-1i) phi T(1-1i)

--- TRUNCATED WORD ---
T(1) phi T(1-1i) phi T(1-1i) phi T(-1-1i) phi T(-3-1i) phi

z = 1.5714285714285714+0i
Continued Fraction Digits: [1; 1-1i, 1-1i, -1-1i, -3-1i]
LaTeX Output: \[1+\frac{1}{1-i+\frac{1}{1-i+\frac{1}{-1-i+\frac
    {1}{-3-i}}}}\]
```

Therefore, we have that

$$\frac{11}{7} = 1 + \frac{1}{1+\frac{1}{1+\frac{1}{3}}} = 1 + \frac{1}{1-i+\frac{1}{1-i+\frac{1}{-1-i+\frac{1}{-3-i}}}}$$

The resulting expansion differs from the classical continued fraction of the same number,

in particular the convergents differ. Despite the construction of the tessellation of $\mathbb{H}^3$ being motivated by that of the tessellation of $\mathbb{H}^2$, this example highlights that the higher-dimensional algorithm encodes different geometric information.

7 Conclusion

Motivated by the classical theory of continued fractions for real numbers, we have developed a geometric algorithm for computing continued fractions of complex numbers via hyperbolic three-space. In this setting, the convergents arise from the intersections of a geodesic with a tessellation by Farey octahedra, directly extending the correspondence between continued fractions and the tessellation by Farey triangles of the hyperbolic plane.

A key feature of this construction is the role of additional symmetries arising in the complex setting. In particular, the appearance of the rotation κ in the decomposition of transformations introduces behaviour not present in the real case. As shown in Lemma 6.6, exploiting the invariance of κ at ∞ allows the continued fraction expansion to be read off from the expansion and resolving an ambiguity that arises in previous approaches such as [Hoc19].

While the algorithm provides a natural geometric generalisation of the real case, its approximation properties were not analysed in full within this work. In particular, obtaining sharp bounds for the error of convergents in the complex setting represents a natural direction for further study.

The ideas developed here also suggest several other avenues for future investigation. One direction is to compare this geometric algorithm with alternative approaches, such as the Hurwitz continued fraction algorithm as described in [Osw14], which imposes a nearest-Gaussian-integer condition and is known to yield strong approximation properties. This algorithm considers the Farey tessellation of $\mathbb{H}^2$ as a restriction in $\mathbb{H}^3$, as opposed to our direction of constructing a new tessellation to use the geometry of the entire space. Understanding the relationship between these methods may provide further insight into the interplay between geometry and Diophantine approximation in $\mathbb{C}$.

More broadly, this framework opens the possibility of extending these ideas to other imaginary quadratic integer rings, or to alternative tessellations of hyperbolic space. Such generalisations could reveal deeper connections between continued fractions, group actions, and the geometry of higher-dimensional hyperbolic spaces.

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8 Appendix

A Code to Compute Real Continued Fractions

```python
import numpy as np

# ==============================
#            GENERATORS
# ==============================

T1 = np.array([[1, 1], [0, 1]])    # x -> x + 1
S = np.array([[0, -1], [1,  0]])   # x -> -1/x

epsilon = 1e-10

# ==============================
#          MOBIUS ACTION
# ==============================

def mobius(M, x):
    a, b, c, d = M.ravel()
    if abs(c*x + d) < epsilon:
        return np.inf
    else:
        return (a*x + b) / (c*x + d)

# ==========================================
#                MAIN ALGORITHM
# ==========================================

def cont_frac_exp_real(x, max_steps = 10):
    M = np.eye(2)
    digits = []
    steps = []

    def record_step():
        steps.append({"M" : M.copy()})

    for _ in range(max_steps):
        if abs(x - round(x)) < epsilon:
            x = round(x)
        if abs(x) < epsilon:
            break

        if x > 0:
            a = int(np.floor(x + epsilon))
```

```
            digits.append(a)

            T_inv_a = np.array([[1, -a], [0,  1]])
            x = mobius(T_inv_a, x)
            M = T_inv_a @ M

            if abs(x) < epsilon:
                record_step()
                break

            x = mobius(S, x)
            M = S @ M
            record_step()

        else:
            a = int(np.floor(-x + epsilon))
            digits.append(a)

            T_a = np.array([[1, a], [0,  1]])
            x = mobius(T_a, x)
            M = T_a @ M

            if abs(x) < epsilon:
                record_step()
                break

            x = mobius(S, x)
            M = S @ M
            record_step()

    return digits, steps

# ==========================================
#            ALLOWS NUMERICAL INPUT
# ==========================================

if __name__ == "__main__":
    user_input = input("Enter a real number (allows for 'np' functions): ")
    steps_input = input("Enter maximum iterations of the algorithm (press Enter for default = 10): ")

    try:
        x = eval(user_input, {"__builtins__": None}, {"np": np})
    except Exception as e:
```

```
        print("Invalid input:", e)
        exit()

    try:
        max_steps = int(steps_input) if steps_input.strip() != ""
            else 10
    except:
        print("Invalid step count, using default = 10")
        max_steps = 10

    if max_steps <= 0:
        print("Steps must be positive, using default = 10")
        max_steps = 10

    digits, steps = cont_frac_exp_real(x, max_steps = max_steps)
    print("x =", x)

    if digits:
      output = "[" + str(digits[0])
      if len(digits) > 1:
          output += "; " + ", ".join(str(d) for d in digits[1:])
      output += "]"
    else:
      output = "[]"

    print("Continued fraction digits:", output)

    array = digits.copy()
    text=''

    def make_frac(t, i, a):
        t=r''+t[:-i]+str(a)+'+\\frac{1}{}'+t[-i:]
        return t

    i = 0
    for a in array:
        text = make_frac(text, i, a)
        i+=1

    text = r'\['+text[:-10-i]+text[-i+1:]+r'\]'
    print(text)
```

B Möbius Extension Computation

$$
\begin{aligned}
\frac{ah+b}{ch+d} &= \frac{a(z+tj)+b}{c(z+tj)+d} \\
&= \frac{(az+b)+atj}{(cz+d)+ctj} \\
&= \frac{((az+b)+atj)(\overline{(cz+d)+ctj})}{|cz+d|^2+|c|^2t^2} \\
&= \frac{((az+b)+atj)((\overline{cz+d})-ctj)}{|cz+d|^2+|c|^2t^2} && \text{(from Lemma 3.1)} \\
&= \frac{(az+b)(\overline{cz+d})-atjctj+atj(\overline{cz+d})-(az+b)ctj}{|cz+d|^2+|c|^2t^2} \\
&= \frac{(az+b)(\overline{cz+d})-at^2jcj+at(cz+d)j-(az+b)ctj}{|cz+d|^2+|c|^2t^2} \\
&= \frac{(az+b)(\overline{cz+d})-at^2j^2\bar{c}+taczj+tadj-taczj-tbcj}{|cz+d|^2+|c|^2t^2} \\
&= \frac{(az+b)(\overline{cz+d})+at^2\bar{c}+t(ad-bc)j}{|cz+d|^2+|c|^2t^2}
\end{aligned}
$$

Writing the j-coefficient separately gives us

$$
\begin{aligned}
\frac{ah+b}{ch+d} &= \frac{(az+b)(\overline{cz+d})+a\bar{c}t^2}{|cz+d|^2+|c|^2t^2} + \frac{t(ad-bc)}{|cz+d|^2+|c|^2t^2}j \\
&= \frac{(az+b)(\overline{cz+d})+a\bar{c}t^2}{|cz+d|^2+|c|^2t^2} + \frac{t}{|cz+d|^2+|c|^2t^2}j
\end{aligned}
$$

C Simplifying Derivative Computations

$$
\begin{aligned}
aQ-cP &= a(|w|^2+|c|^2t^2)-c((az+b)\bar{w}+a\bar{c}t^2) \\
&= a|w|^2-c(az+b)\bar{w} \\
&= \bar{w}(aw-c(az+b)) \\
&= \bar{w}(a(cz+d)-c(az+b)) \\
&= \bar{w}(ad-bc) \\
&= \bar{w}
\end{aligned}
$$

$$
\begin{aligned}
(az+b)Q - wP &= (az+b)(|w|^2 + |c|^2 t^2) - w((az+b)\bar{w} + a\bar{c}t^2) \\
&= (az+b)|c|^2 t^2 - wa\bar{c}t^2 \\
&= \bar{c}t^2((az+b)c - wa) \\
&= \bar{c}t^2(acz + bc - acz - ad) \\
&= -\bar{c}t^2
\end{aligned}
$$

D Zero Entries of $J^T J$

$$
\begin{aligned}
\langle Df_g{}^{(x)}, Df_g{}^{(t)} \rangle &= \mathfrak{Re}(\partial_x z' \overline{\partial_t z'}) + \partial_x t' \partial_t t' \\
&= \mathfrak{Re}\left(\frac{\bar{w}^2 - \bar{c}^2 t^2}{Q^2} \cdot \frac{2ctw}{Q^2} \right) - \frac{(\bar{c}w + c\bar{w})t}{Q^2} \cdot \frac{|w|^2 - |c|^2 t^2}{Q^2} \\
&= \frac{1}{Q^4} \mathfrak{Re}(2c\bar{w}^2 wt - 2\bar{c}^2 cwt^3) - \frac{1}{Q^4}(2\ \mathfrak{Re}(c\bar{w}))t \underbrace{(|w|^2 - |c|^2 t^2)}_{\in \mathbb{R}} \\
&= \frac{1}{Q^4} \mathfrak{Re}(2c\bar{w}|w|^2 t - 2\bar{c}|c|^2 wt^3) - \frac{1}{Q^4} \mathfrak{Re}(2c\bar{w}|w|^2 t - 2\bar{c}|c|^2 wt^3) \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
\langle Df_g{}^{(y)}, Df_g{}^{(t)} \rangle &= \mathfrak{Re}(\partial_y z' \overline{\partial_t z'}) + \partial_y t' \partial_t t' \\
&= \mathfrak{Re}\left(i\frac{\bar{w}^2 + \bar{c}^2 t^2}{Q^2} \cdot \frac{2ctw}{Q^2} \right) + i\frac{(\bar{c}w - c\bar{w})t}{Q^2} \cdot \frac{|w|^2 - |c|^2 t^2}{Q^2} \\
&= -\frac{1}{Q^4} \mathfrak{Im}\left((\bar{w}^2 + \bar{c}^2 t^2) \cdot 2ctw \right) + \frac{1}{Q^4} \cdot 2t\ \mathfrak{Im}(c\bar{w})(|w|^2 - |c|^2 t^2) \\
&= -\frac{1}{Q^4} \mathfrak{Im}(2c\bar{w}|w|^2 t + 2\bar{c}|c|^2 wt^3) + \frac{1}{Q^4} \mathfrak{Im}(2c\bar{w}|w|^2 t - 2c|c|^2 \bar{w}t^3) \\
&= -\frac{1}{Q^4} \mathfrak{Im}(2c\bar{w}|w|^2 t + 2\bar{c}|c|^2 wt^3) + \frac{1}{Q^4} \mathfrak{Im}(2c\bar{w}|w|^2 t + 2\bar{c}|c|^2 \bar{w}t^3) \\
&= 0
\end{aligned}
$$

E Diagonal Entries of $J^T J$

$$
\begin{aligned}
|Df_g{}^{(y)}|^2 &= \langle Df_g{}^{(y)}, Df_g{}^{(y)} \rangle \\
&= |\partial_y z'|^2 + (\partial_y t')^2 \\
&= \frac{1}{Q^4}|\bar{w}^2 + \bar{c}^2 t^2|^2 - \frac{1}{Q^4}(\bar{c}w - c\bar{w})^2 t^2 \\
&= \frac{1}{Q^4}(|w|^4 + |c|^4 t^4 + 2\mathfrak{Re}(\bar{w}^2 \overline{\bar{c}^2 t^2}) - \bar{c}^2 w^2 t^2 + c^2 \bar{w}^2 t^2 + 2|c|^2 |w|^2 t^2) \\
&= \frac{1}{Q^4}(|w|^4 + |c|^4 t^4 + 2\mathfrak{Re}(c^2 \bar{w}^2 t^2) - 2\mathfrak{Re}(c^2 \bar{w}^2 t^2) + 2|c|^2 |w|^2 t^2) \\
&= \frac{1}{Q^4}(|w|^2 + |c|^2 t^2)^2 \\
&= \frac{1}{Q^2}
\end{aligned}
$$

$$
\begin{aligned}
|Df_g{}^{(t)}|^2 &= \langle Df_g{}^{(t)}, Df_g{}^{(t)} \rangle \\
&= |\partial_t z'|^2 + \partial_t t'^2 \\
&= \frac{1}{Q^4} \cdot 4|c|^2 |w|^2 t^2 + \frac{1}{Q^4}(|w|^4 + |c|^4 t^4 - 2|c|^2 |w|^2 t^2) \\
&= \frac{1}{Q^4}(|w|^4 + |c|^4 t^4 + 2|c|^2 |w|^2 t^2) \\
&= \frac{1}{Q^4}(|w|^2 + |c|^2 t^2)^2 \\
&= \frac{1}{Q^2}
\end{aligned}
$$

F Relation Computations

Firstly, from $(ST_1)^3 = I, S^2 = I$, we have

$$
\begin{aligned}
(ST_1^{-1})^3 &= S(T_1^{-1} S T_1^{-1} S T_1^{-1}) \cdot S^2 \\
&= S(ST_1)^{-3} S \\
&= S^2 \\
&= I
\end{aligned}
$$

From $(T_1 R)^2 = I$ and $R^2 = I$, we have

$$
\begin{aligned}
(T_1^{-1} R)^2 &= T_1^{-1} R T_1^{-1} R \\
&= T_1^{-1} R (R T_1 R) R \\
&= T_1^{-1} T_1 \\
&= I
\end{aligned}
$$

From $R^2 = I$ and $[T_1, T_i] = I$, we have

$$\begin{aligned}(T_1T_iR)^2 &= T_1T_iRT_1T_iR \\ &= T_1T_i(RT_1R)(RT_iR) \\ &= T_1T_iT_1^{-1}T_i^{-1} \\ &= I\end{aligned}$$

and also

$$\begin{aligned}(T_1^{-1}T_iR)^2 &= T_1^{-1}T_iRT_1^{-1}T_iR \\ &= T_1^{-1}T_i(RT_1^{-1}R)(RT_iR) \\ &= T_1^{-1}T_iT_1T_i^{-1} \\ &= I\end{aligned}$$

G Chapter 6 Stuff

Define the elements

$$g_1 = I,\ g_2 = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix},\ g_3 = \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix},\ g_4 = \begin{pmatrix} 1 & -1+i \\ 0 & i \end{pmatrix}.$$

These allow us to define the vertex sets of the vertical faces of $\mathbb{O}_0$ i.e. those sharing a vertex at ∞, which are given by $g_1\mathbb{T}_0 = \{\infty, 0, 1\}$, $g_2\mathbb{T}_0 = \{\infty, i, 1+i\}$, $g_3\mathbb{T}_0 = \{\infty, 0, i\}$, and $g_4\mathbb{T}_0 = \{\infty, 1+i, 1\}$. We then look at the other 4 faces. Here, the geodesic algorithm deviates from the classical case as in $\mathbb{H}^2$ because a geodesic translated into $\mathbb{O}_0$ has 4 potential faces to cross. Therefore, we must specify what happens at each.

Let T_α be the translation by $\alpha = a + bi$ for $a, b \in \mathbb{Z},\ b \geq 0$. We can restrict b to the non-negative integers by invoking the rotation g_3, and otherwise taking inverses of the translation. We want to write the translation T_α as a composition of g_2, g_3, and g_4 with non-negative exponents. Ensuring non-negative exponents ensures that we don't cross backwards through a face we already crossed. This reflects the nature of the geodesic.

In particular, given $g_3T_\alpha g_3^{-1} = T_{i\alpha}$ and that $g_3^4 = I$, we have

- For $a \leq 0 \leq b$,
$$T_\alpha = T_1^aT_i^b = T_{-1}^{-a}T_i^b = g_3T_i^{-a}g_3^{-1}T_i^b$$
with $-a, b \geq 0$. Therefore, $T_\alpha = g_3g_2^{-a}g_3^3g_2^b$
- For $0 \leq a \leq b$,
$$T_\alpha = T_1^aT_i^b = T_1^a(T_i^aT_i^{-a})T_i^b = T_{1+i}^aT_i^{b-a}$$

with $a, b-a \geq 0$. Furthermore,

$$g_4 g_3 = \begin{pmatrix} 1 & -1+i \\ 0 & i \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} i & -1+i \\ 0 & i \end{pmatrix} \sim \begin{pmatrix} 1 & 1+i \\ 0 & 1 \end{pmatrix}$$

Therefore, $T_\alpha = (g_4 g_3)^a g_2^{b-a}$

- For $0 \leq b \leq a$,

$$T_\alpha = T_i(b - ia) = g_3 T_{b-ia} g_3^{-1} = g_3 T_{b(1+i)-(a+b)i} g_3^{-1} = g_3 T_{1+i}^b T_i^{-(a+b)} g_3^{-1}$$

Since $g_3^2 T_\alpha g_3^{-2} = T_{-\alpha}$ we have that

$$T_\alpha = g_3 T_{1+i}^b g_3^2 T_i^{a+b} g_3^{-3} = g_3 T_{1+i}^b g_3^2 T_i^{a+b} g_3$$

with $b, a+b \geq 0$. Therefore, $T_\alpha = g_3 (g_4 g_3)^b g_3^2 g_2^{a+b} g_3$

The key idea is that, by writing T_α in terms of g_i with positive powers, we fix the direction of the geodesic algorithm. Each step corresponds to moving from one Farey octahedron in the tessellation to the next, and positive powers of the g_i guarantees the algorithm keeps moving forward, preventing any steps being "undone" and cutting back through a previously met octahedron.

H Approximation Stuff

We now look at the application of this algorithm to approximation.

####### Approximating Complex Numbers using the Geodesic Algorithm

We now look at the accuracy of the nth convergents as an approximation to $z \in \mathbb{C}$ by this algorithm. The following is analogous to convergence in the 2-dimensional case given in Theorem 2.7.

Before we look at this, we need some more information about the Farey triangles. In [Hoc19], Hockman defines the *norm* of a triangle $\mathbb{T} = \{\alpha_i/\gamma_i : i = 1, 2, 3\}$ to be

$$N(\mathbb{T}) = \sum_{i=1}^{3} N(\gamma_i)$$

As well as this, we will define $S_\mathbb{T}$ to be the circumsphere (hemisphere) orthogonal to $\widehat{\mathbb{C}}$ defined by the 3 points α_i/γ_i. Using these definitions, we will state without proof one of the main results from this paper.

Theorem 8.1. *Every geodesic in $\mathbb{H}^3$ ending at $w = \xi$ cuts through a chain of hyperbolic Farey triangles, $\{\mathbb{T}_k\}$ for $k \in \mathbb{N}$, with each Farey triangle in the inner subdivision of its predecessor, and with $N(\mathbb{T}_{j+1}) > N(\mathbb{T}_j)$. Further, $N(\mathbb{T}_k) \to \infty$ as $k \to \infty$ while the*

vertices of the Farey triangles $\mathbb{T}_k$ tend to ξ as $k \to \infty$.

This brings us to our result about the convergents.

Theorem 8.2. *Let $z = [\alpha_0; \alpha_1, \alpha_2, ...] \in \mathbb{C}$ be a Gaussian integer continued fraction with $\alpha_n \in \mathbb{Z}[i]$. Let*

$$\frac{p_n}{q_n} = [\alpha_0; \alpha_1, \alpha_2, ..., \alpha_n]$$

be the convergents. Then

1) $p_n q_{n+1} - p_{n+1} q_n \in \mathbb{Z}[i]^\times$

2) $p_{n+1} = \alpha_{n+1} p_n + p_{n-1}$ *and* $q_{n+1} = \alpha_{n+1} q_n + q_{n-1}$, *where* $p_{-1} = 1,\ q_{-1} = 0$.

3) *The convergents occur as vertices of a nested sequence of Farey triangle, whose diameters tend to 0.*

Proof. We can represent a single stage s_n of the algorithm by a product of matrices $s_n = \phi T_{\alpha_n} \kappa^r$. From Lemma 6.6, we are able to reduce this to $s_n = \phi T_{\alpha_n}$ by "sifting" each κ to the end of the transformation. However, $\kappa^r \cdot \infty = \infty$, thus since we are concerned with a geodesic joining ∞ to z, we can without loss of generality exclude the κ^r from M_n.

We now define the matrix $M \in PSL_2(\mathbb{Z}[i])$ to be $M_n = T_{\alpha_0} \phi T_{\alpha_1} \phi ... \phi T_{\alpha_n} \phi \kappa^r$. Observe that $M^{-1} = \kappa^r \phi T_{-\alpha_n} \phi T_{-\alpha_{n-1}} \phi ... \phi T_{-\alpha_1} \phi T_{-\alpha_0}$.

$$\begin{aligned} M_n^{-1}[a_0; a_1, a_2, ..., a_n] &= (\phi T_{-\alpha_n} \phi T_{-\alpha_{n-1}} \phi ... \phi T_{-\alpha_1} \phi T_{-\alpha_0})[a_0; a_1, a_2, ..., a_n] \\ &= (\phi T_{-\alpha_n} \phi T_{-\alpha_{n-1}} \phi ... \phi T_{-\alpha_1} \phi)[0; \alpha_1, \alpha_2, ..., \alpha_n] \\ &= (\phi T_{-\alpha_n} \phi T_{-\alpha_{n-1}} \phi ... \phi T_{-\alpha_1})[\alpha_1; \alpha_2, ..., \alpha_n] \end{aligned}$$

As in the proof of Theorem 2.7, we have the same result without needing to consult even and odd cases individually. That is,

$$M_n^{-1}[\alpha_0; \alpha_1, ..., \alpha_n + \xi] = \frac{1}{\xi}$$

Therefore, evaluating at $\xi = 0$ and $\xi = \infty$,

$$M_n^{-1}\left(\frac{p_n}{q_n}\right) = \infty, \qquad M_n^{-1}\left(\frac{p_{n-1}}{q_{n-1}}\right) = 0$$

Thus,

$$M_n = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \Longrightarrow \det \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = 1 \Longrightarrow \det \begin{pmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{pmatrix} = -1$$

Thus since the matrix is equivalent up to multiplication by some κ^r, $p_{n-1} q_n - p_n q_{n-1} \in$

$\mathbb{Z}[i]^\times$ for all $n \in \mathbb{N}$, proving *1)*.

The proof for *2)* is equivalent to that in Theorem 2.7.

For *3)*, we refer to Theorem 8.1. By this theorem, the geodesic joining ∞ to z passes through a nested sequence of Farey triangles $\{\mathbb{T}_k\}_k$ satisfying $N(\mathbb{T}_k) \to \infty$ as $k \to \infty$, each of which contains z. The diameters of the circumspheres of these triangles therefore tend to 0, and their vertices converge to z.

□

Corollary 8.3. *With the notation as in Theorem 8.2, $p_n/q_n \to z$ as $n \to \infty$. Moreover, we can bound $|\xi - p_n/q_n|$ by a telescoping sum*

$$\left| z - \frac{p_n}{q_n} \right| \leq \sum_{k=n}^{\infty} \frac{1}{|q_k q_{k+1}|}$$

Proof. From Theorem 8.2, it follows that $p_n/q_n \to z$ since the convergents p_n/q_n appear as vertices of nested Farey triangles. It follows that

$$\left| z - \frac{p_n}{q_n} \right| \leq \sum_{k=n}^{\infty} \left| \frac{p_{k+1}}{q_{k+1}} - \frac{p_k}{q_k} \right| = \sum_{k=n}^{\infty} \left| \frac{p_{k+1}q_k - p_k q_{k+1}}{q_k q_{k+1}} \right| = \sum_{k=n}^{\infty} \frac{1}{|q_k q_{k+1}|}$$

□

In the real case, we could show that $|q_{n+1}| \geq |q_n|$ and $|q_n| \geq n$ for all $n \in \mathbb{N}$ and thus show that the error of the nth convergent as an approximation was bounded above by $1/n(n+1)$. However, it is difficult to find an easily calculable error bound for the GICF. To achieve something similar, we would need to ensure that $|\alpha_{n+1}| \geq 2$. Indeed, $q_1 = \alpha_1 q_0 + q_{-1} = \alpha_1 q_0$, so $|q_1| \geq |q_0|$, and an inductive argument shows

$$\begin{aligned} |q_{n+1}| &= |\alpha_{n+1} q_n + q_{n-1}| \\ &\geq ||\alpha_{n+1}||q_n| - |q_{n-1}|| \\ &\geq (|\alpha_{n+1}| - 1)|q_n| \end{aligned}$$

This is not a condition we can guarantee, namely with translations T_1 and T_i. While the algorithm exploits the geometry of $\mathbb{H}^3$ to evaluate continued fractions, calculating the error bound is not as simple as in the real case.

Considering the bound in Corollary 8.3, we note that

$$\left| \frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n} \right| = \frac{1}{|q_n q_{n+1}|},$$

which is much easier computation and represents the distance between two consecutive

convergents. It therefore provides a reasonable scale for the approximate error, avoiding the computational burden of the original infinite series. In the following example, we will compare the error with this term to illustrate the rate at which convergents approach z.

Example 8.4. Let $z = \dfrac{1+2i+\sqrt{1+4i}}{2}$.

1) We define $z_1 = T_{1+i}^{-1} \cdot z = \dfrac{-1+\sqrt{1+4i}}{2} \in f_2\mathbb{T}_0$.

2) In fact, a quick computation shows that $T_1^{-1} f_2^{-1} T_{1+i}^{-1}(z) = z_1$, so z_1 is a fixed point of the transformation $T_1^{-1} f_2^{-1}$.

So

$$\frac{1+2i+\sqrt{1+4i}}{2} = [1+i, -i, 1, -i, 1, ...] = 1 + i + \frac{1}{-i + \frac{1}{1+\frac{1}{-i+\frac{1}{1+...}}}}$$

Observe that $|\alpha_n| < 2$, so we are unable to conclude that $|q_n| \leq |q_{n+1}|$ for an accurate error bound, so we now look at the error estimate provided by the leading term of the infinite series bound. Then using the recurrence relation for both p_n and q_n, we get

$$\frac{p_1}{q_1} = \frac{2-i}{-i}, \qquad \frac{p_2}{q_2} = \frac{3}{1-i}, \qquad \frac{p_3}{q_3} = \frac{2-4i}{-1-2i}, \qquad \frac{p_4}{q_4} = \frac{5-4i}{-3i}, \qquad \frac{p_5}{q_5} = \frac{-2-9i}{-4-2i}$$

	$\frac{p_0}{q_0}$	$\frac{p_1}{q_1}$	$\frac{p_2}{q_2}$	$\frac{p_3}{q_3}$	$\frac{p_4}{q_4}$
$\frac{1}{\lvert q_n q_{n+1}\rvert}$	1	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{10}}$	$\frac{1}{3\sqrt{5}}$	$\frac{1}{6\sqrt{5}}$
$\left\lvert \xi - \frac{p_n}{q_n} \right\rvert$	0.4805	0.2309	0.0554	0.0107	0.0028
Error	0.5195	0.476207	0.2608	0.1384	0.0717

Table 1: Approximating the Error of the First Few Convergents

This shows that the naïve bound we chose is not practical for what we require, nor even the first term of the series.

Although the geodesic algorithm described above produces a Gaussian integer continued fraction expansion for any complex number, it is not very efficient. The present algorithm does not enforce a 'nearest–integer' condition, so the resulting convergents need not satisfy comparably sharp approximation bounds. Nevertheless, the geodesic algorithm has significant advantages as a result of its geometric nature. In particular, it provides a natural interpretation of Gaussian continued fractions in terms of the action of $PSL_2(\mathbb{Z}[i])$

(and by extension $PGL_2(\mathbb{Z}[i])$ with the rotation κ) and the Farey tessellation of H^3, with convergents appearing as vertices of nested Farey triangles intersected by a geodesic. This geometric viewpoint connects the theory of continued fractions with hyperbolic geometry, Möbius transformations, and group actions, offering structural insight.